



# **Experimental methods in solid-state physics**

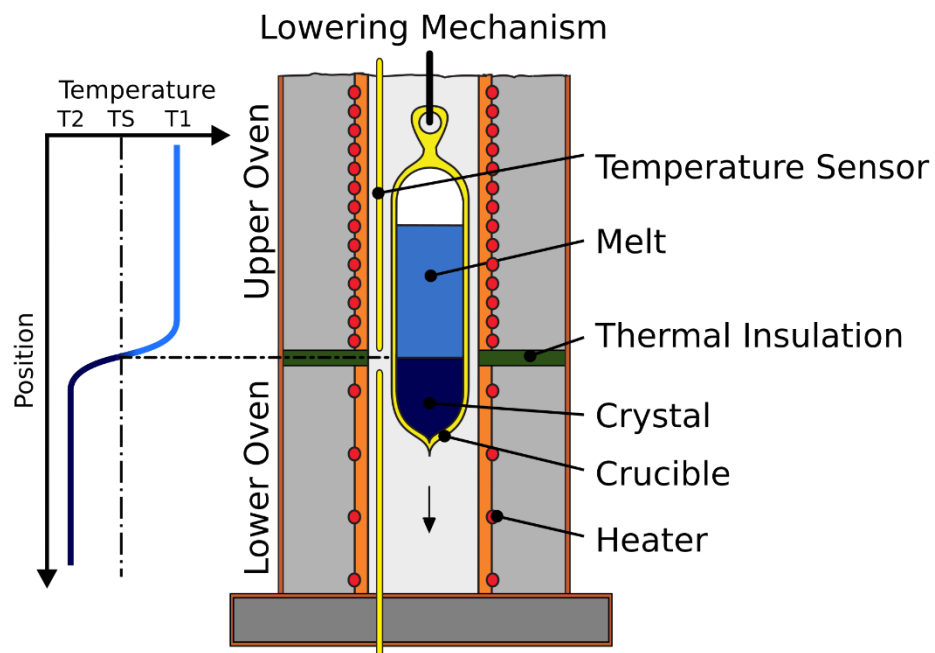
## **01 Crystal growth**

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Walter Schottky Institut and Physics Department  
Zentrum für Nanotechnologie  
und Nanomaterialien

## Czochralski method (pulling from the melt)

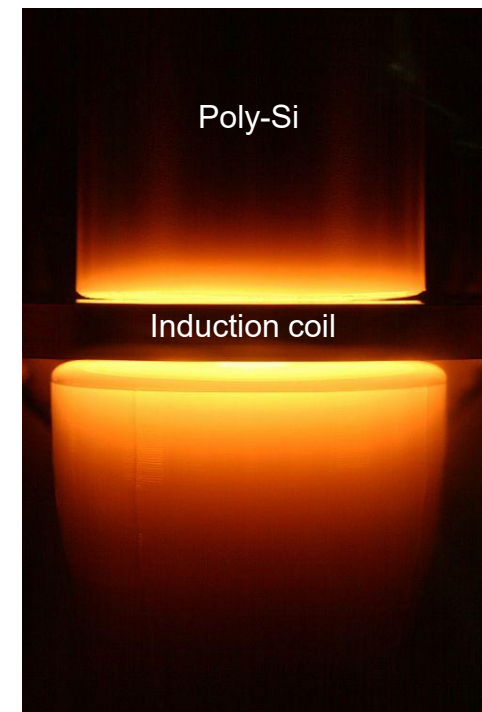


## Bridgman method (temperature gradient)

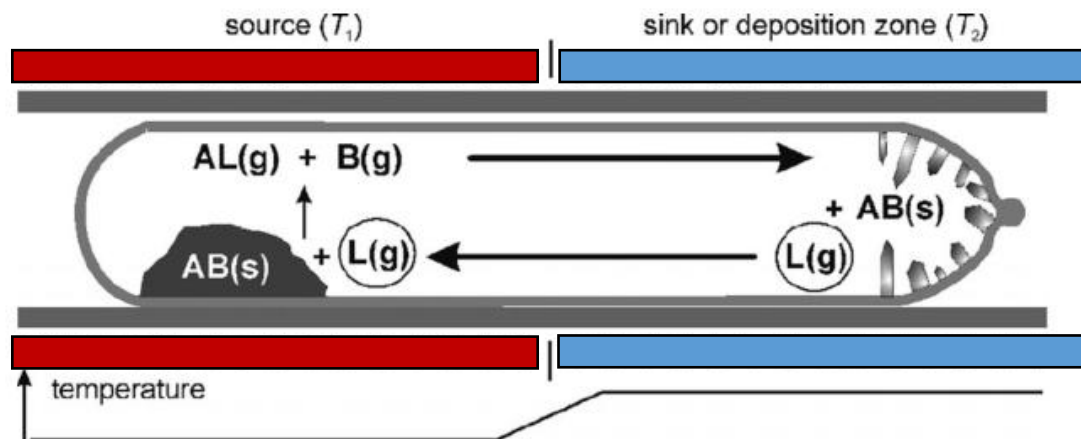


Wikipedia, Cepheiden CC BY-SA 3.0.

## Float zone (inductive melting)



Wikipedia, Matthias Renner CC BY-SA 3.0.



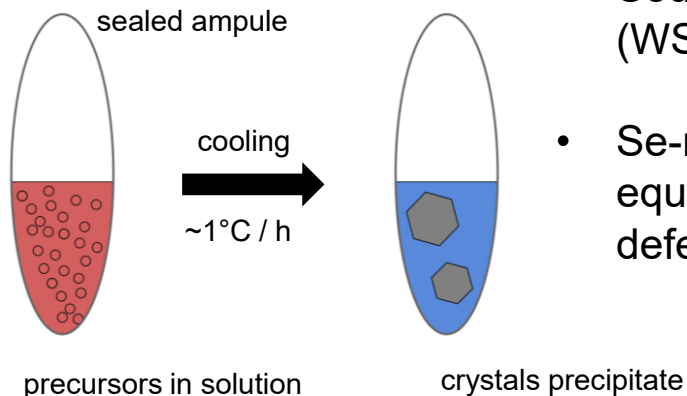
**Figure 2.** Sketch of a transport ampoule in the temperature gradient of a two-zone tubular furnace. In an exothermic reaction solid  $AB$  and the transport agent  $[L(g)]$ , which might be regarded as solvent] form the transporting gas species  $AL(g)$  and  $B(g)$ . At the sink, which is for the chosen example at the higher temperature, the reaction is reversed.

Binnewies *et al.* Z. Anorg. Allg. Chem. 2013, 639, (2), 219–229, Copyright Wiley 2013.

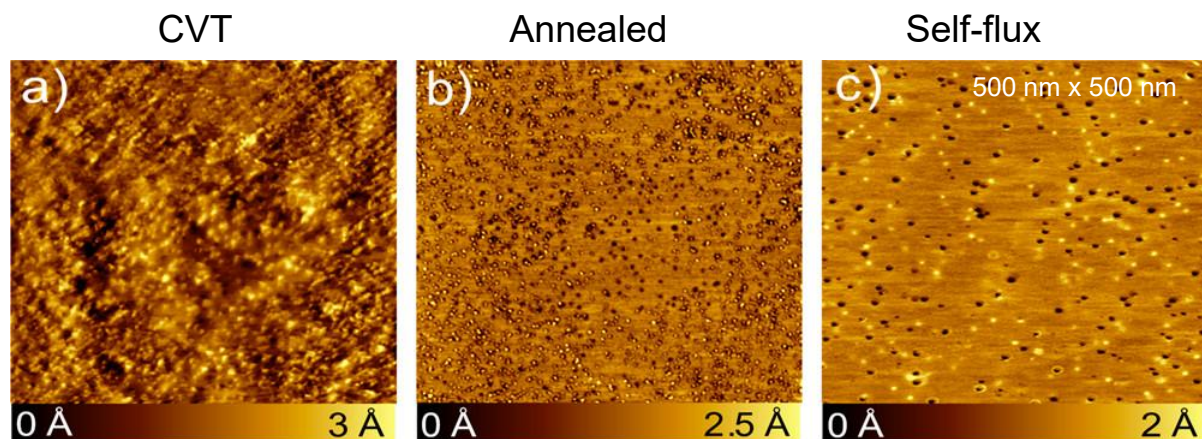


- Transport reaction of the form
  1.  $AB(s) + X(g) \rightarrow AX(g) + B(g)$
  2. Reversed reaction
- $X(g)$  forms a volatile component with the source material  $AB(s)$ . It is called the transport agent.
- The reactants sealed in the reactor (e.g. glass ampule) and placed in a temperature gradient.
- The growth process is completely controlled by thermodynamics → highly repeatable and predictable.
- High quality bulk single crystals can be fabricated from lower quality source materials, e.g. powder.
- Materials exploration → New materials can be rapidly synthesized and their structural and chemical properties can be characterized (X-ray crystallography, XPS analysis, etc.)

## Self-flux growth of WSe<sub>2</sub>



- Source material is dissolved (WSe<sub>2</sub>) in a solvent (Se)
- Se-rich conditions reduce the equilibrium concentrations of Se-defects



Nano Lett. 2019, 19, 7, 4371–4379, <https://doi.org/10.1021/acs.nanolett.9b00985>

## Defect density in equilibrium

- The equilibrium defect concentration is given by the Arrhenius equation

$$n = N_0 \cdot e^{-E/(k_b T)}$$

- $E$ : formation energy of the defect (in eV)

### Example:

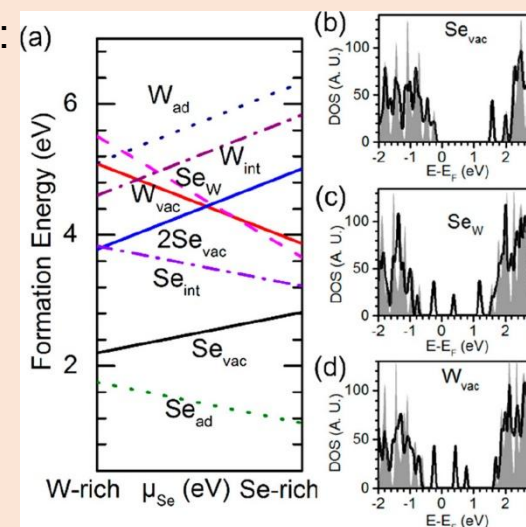
- For  $E = 2$  eV,  $T = 1400$  K:

$$\frac{n}{N} \approx e^{-1/(8.6e-5 \cdot 1400)} \approx e^{-16} \approx 0.1 \text{ ppm}$$

- For  $E = 4$  eV,  $T = 1400$  K:

$$\frac{n}{N} \approx e^{-32} \approx 10 \text{ ppt}$$

ACS Nano 2019, 13, 5, 6050–6059  
<https://doi.org/10.1021/acs.nano.9b02316>



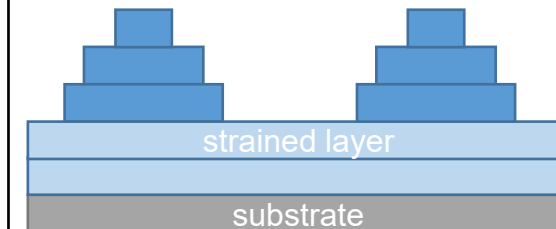
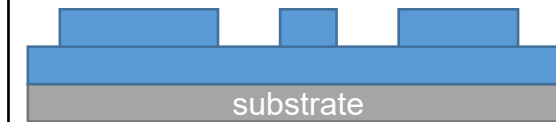
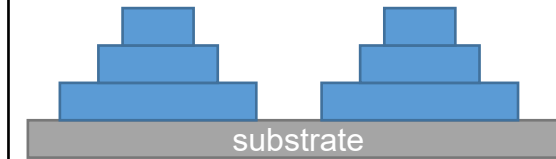


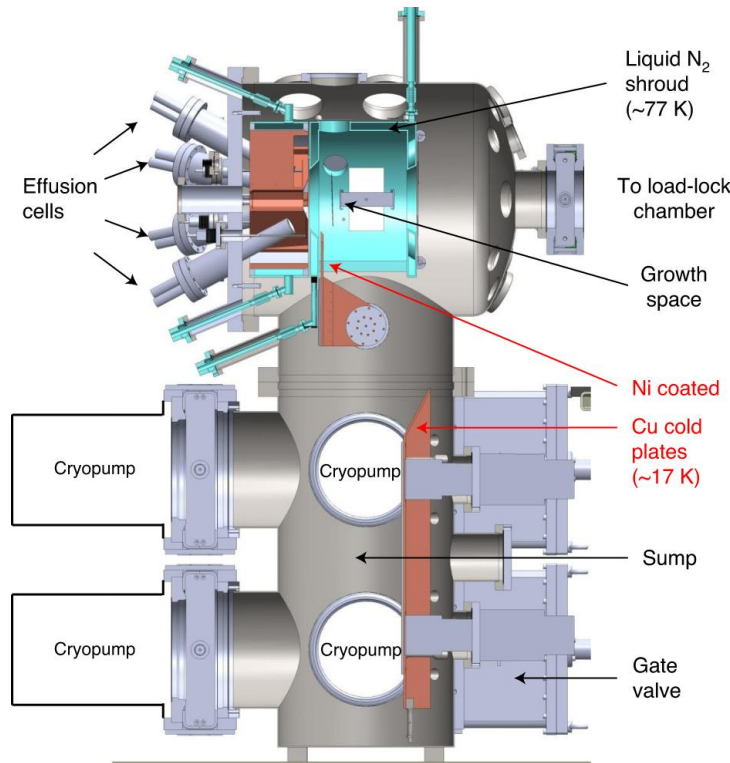
# ***Defect density in equilibrium***



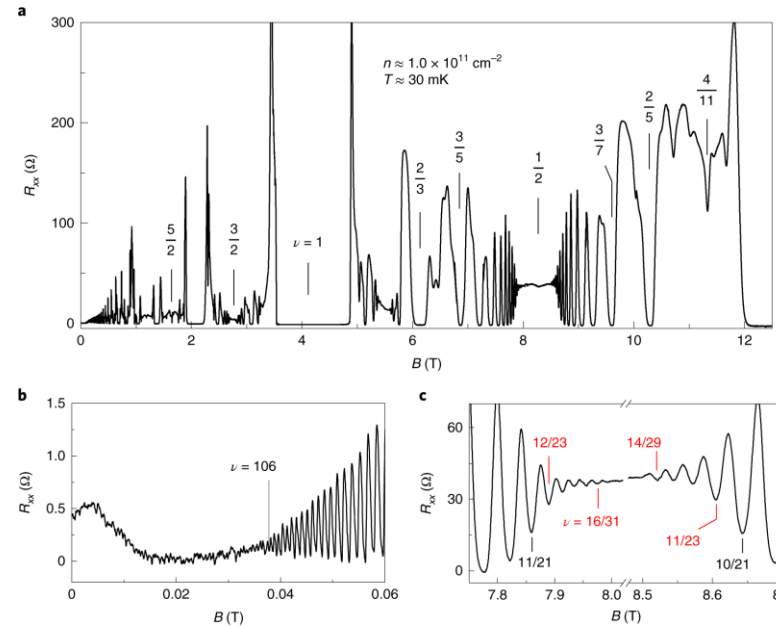


| Growth Mode                | Description                                                                    | Driving Force                               | Morphology                       | Typical Conditions / Examples                         |
|----------------------------|--------------------------------------------------------------------------------|---------------------------------------------|----------------------------------|-------------------------------------------------------|
| <b>Volmer–Weber</b>        | Island growth: atoms bond more strongly to each other than to the substrate.   | High adatom–adatom binding energy           | 3D islands                       | Metals on insulators; e.g. Au on SiO <sub>2</sub>     |
| <b>Frank–van der Merwe</b> | Layer-by-layer growth: atoms bond more strongly to the substrate.              | High adatom–substrate binding energy        | Uniform monolayers               | Epitaxial semiconductors with good lattice match      |
| <b>Stranski–Krastanov</b>  | Layer-plus-island growth: initial layer-by-layer followed by island formation. | Strain accumulation due to lattice mismatch | Monolayer followed by 3D islands | Ge on Si, InAs on GaAs (strained-layer heteroepitaxy) |



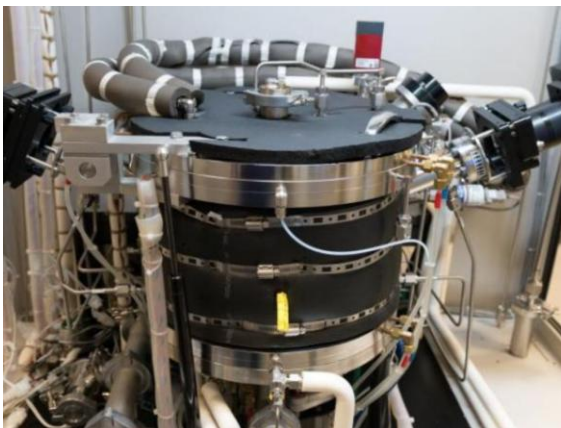


Nature Materials 20, 632–637 (2021)

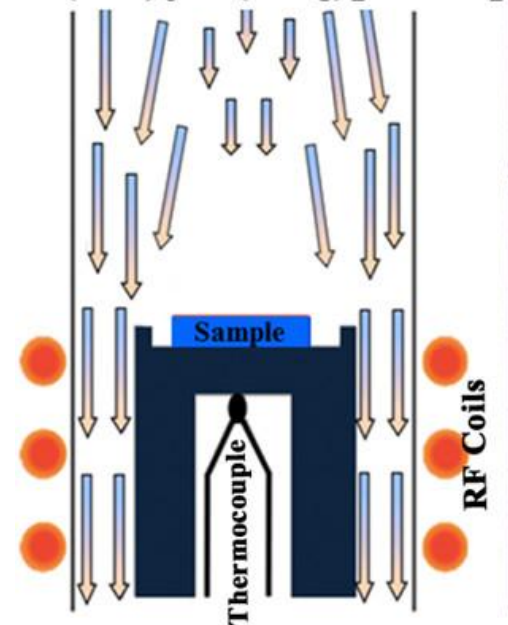


- Material growth directly from the corresponding elements.
- Ultra-high-vacuum environment ( $p = 10^{-10}$  mbar)
- Epitaxial growth. Vertical growth rate  $\sim \text{\AA}/\text{min}$ . Suitable for thin films.
- Technologically very relevant, since it provides very high material quality and is also scalable.

MOCVD Reactor



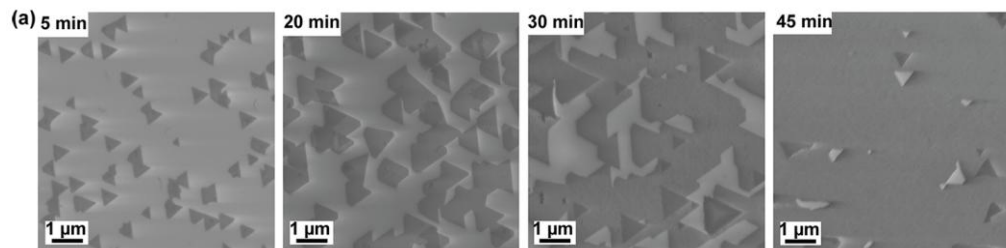
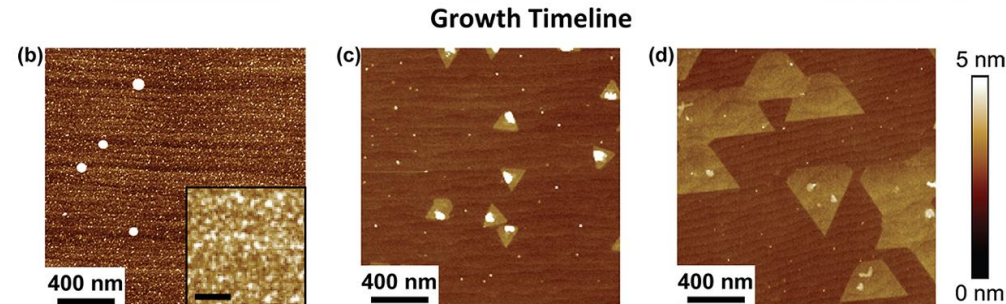
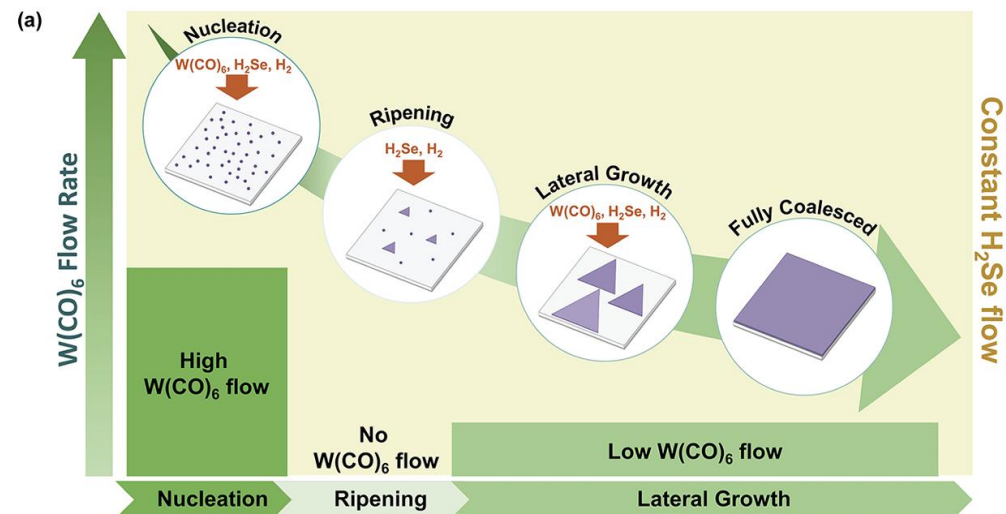
Gas flow control



- Metal organic precursors
- High-vacuum environment
- Precise control over temperature, flow rate, flow profile.
- Technologically very relevant, since it provides very high material quality and is also scalable (e.g. LEDs)

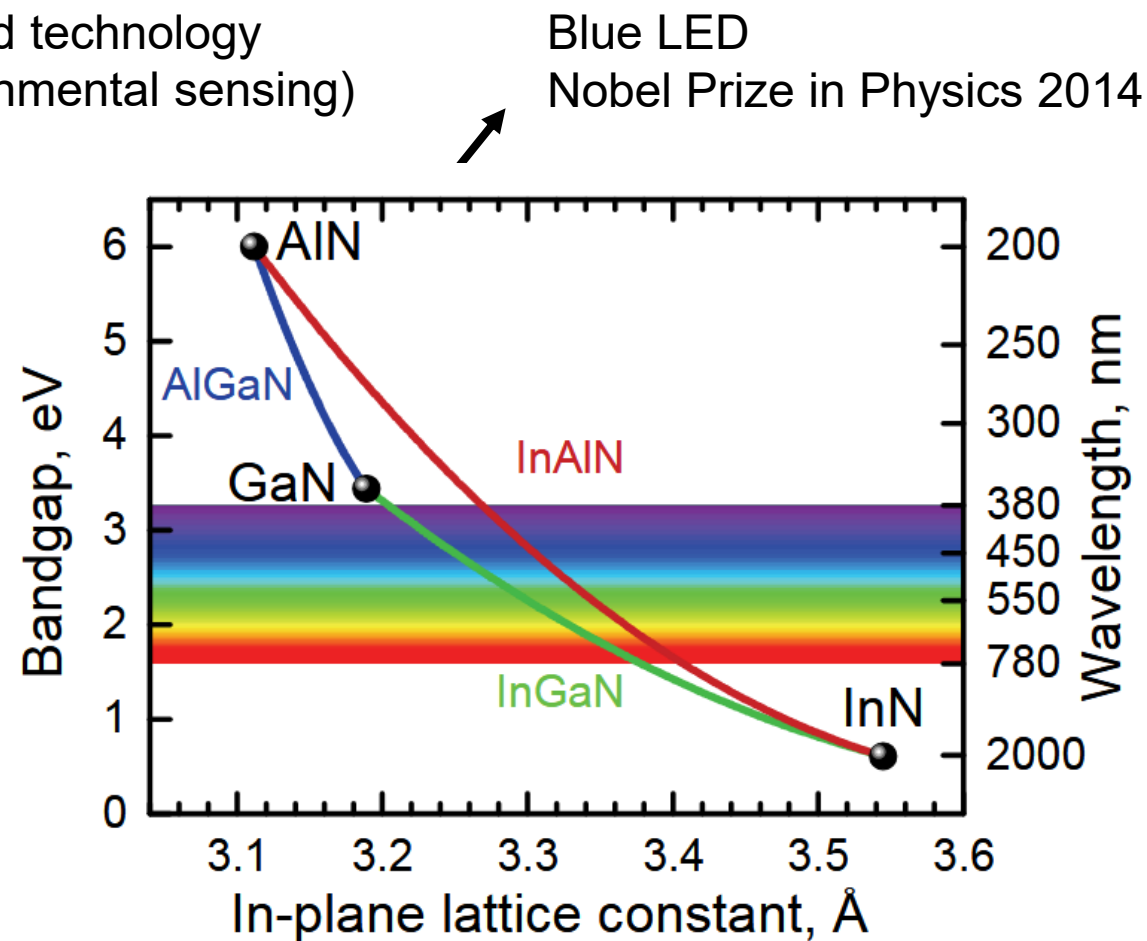
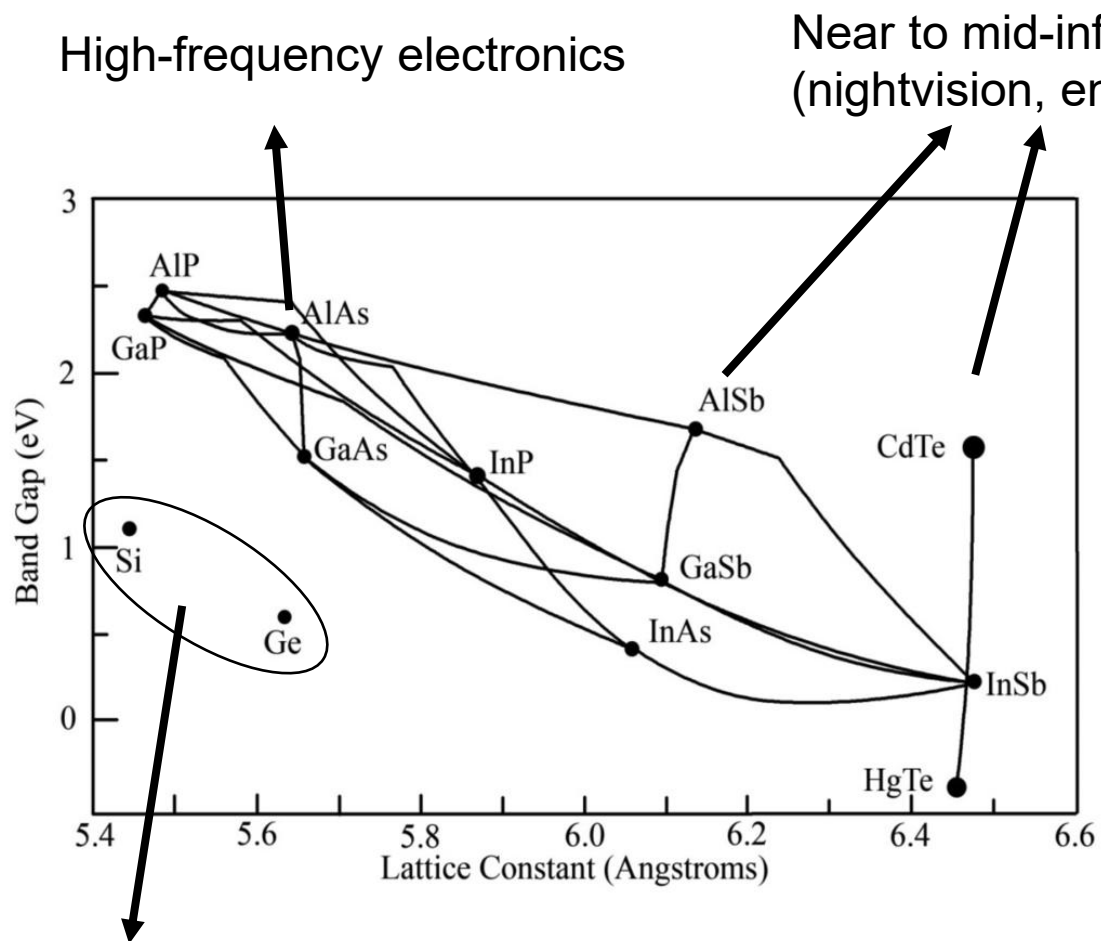
[ACS Nano 9, 2, 2080–2087 \(2015\).](#)

[Link](#)



[Nano Lett. 18, 1049–1056 \(2018\)](#)





**Lattice matching between materials with vastly different lattice constants**

target film



+



substrate

???

Lattice matching between materials with vastly different lattice constants

target film



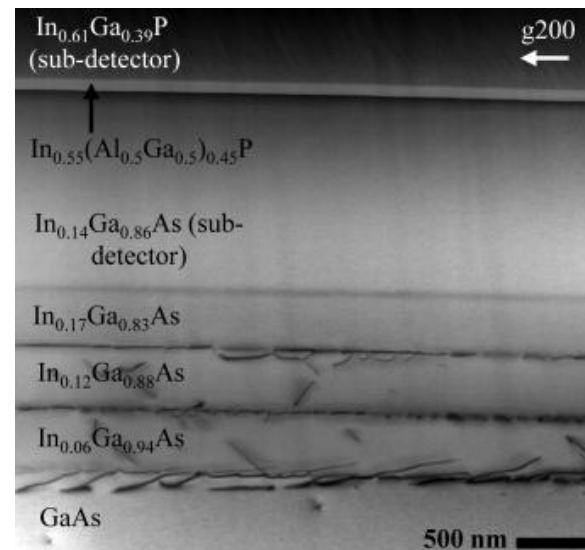
+



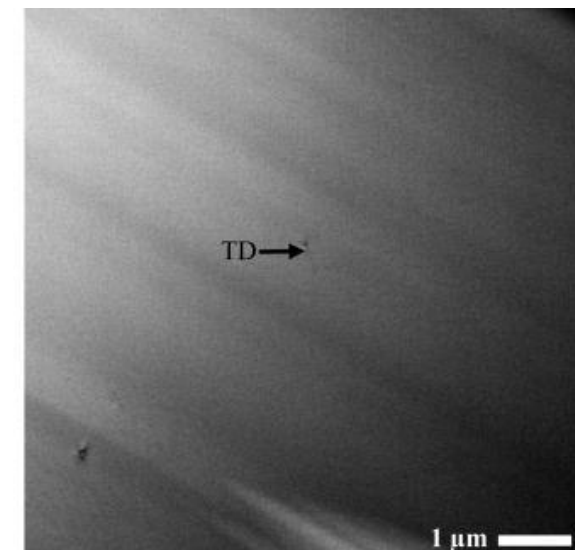
substrate

???

Side view

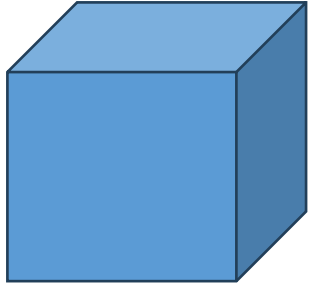


Top view



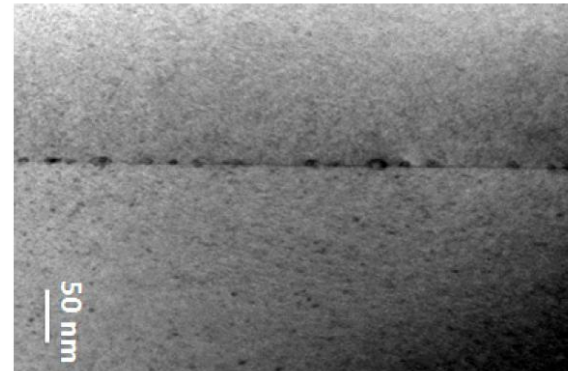
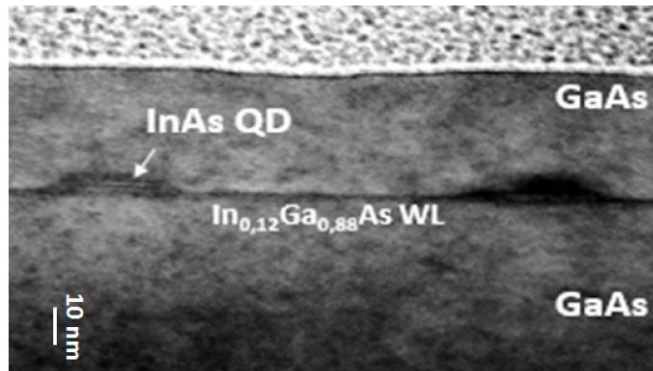
- Lattice matching by growth of a thick buffer layer
- Strain relaxation and defect formation inside the buffer layer
- Graded buffer: stepwise or continuous adaption to target lattice → interfaces or buffer layer traps the defects (mainly threading dislocations → „durchlaufende Versetzung“)

Quantum dot = particle in a box



Energy quantization in all spatial dimensions:

- 0D electron system (artificial atom)
- Discrete energy states



- Growth of a thin strained layer on a „wetting layer“
- Beyond critical thickness → island formation (Stranski-Krastanoff growth)
- Overgrowth of the islands → self-assembled embedded quantum dots

→ see for example lecture on **Advanced Semiconductor Physics**

[Nanomaterials 2020, 10\(12\), 2450;](#)





# **Experimental methods in solid-state physics**

## **02 Real space structure – STM & AFM**

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# Nobel Prize in Physics 1986



Photo: Marianne Kischke, Fritz Haber Institut der Max Planck Gesellschaft. Nobel Foundation archive.

Ernst Ruska

Prize share: 1/2

Fritz-Haber-Institut (Berlin)



Photo from the Nobel Foundation archive.

Gerd Binnig

Prize share: 1/4



Photo from the Nobel Foundation archive.

Heinrich Rohrer

Prize share: 1/4

IBM (Rüschlikon)

The Nobel Prize in Physics 1986 was divided, one half awarded to Ernst Ruska "for his fundamental work in electron optics, and for the design of the first electron microscope", the other half jointly to Gerd Binnig and Heinrich Rohrer "for their design of the scanning tunneling microscope"

To cite this section

MLA style: The Nobel Prize in Physics 1986. NobelPrize.org. Nobel Prize Outreach 2025. Mon. 27 Oct 2025. <<https://www.nobelprize.org/prizes/physics/1986/summary/>>

## Transmission electron microscope 1931

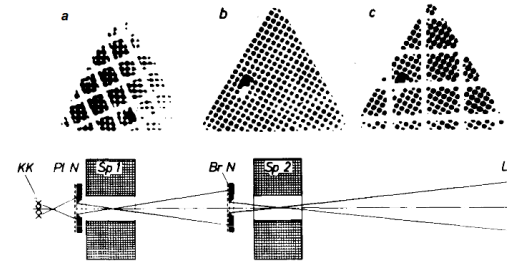
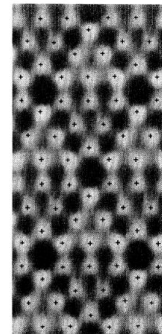


Fig. 3: First experimental proof (7 April 1931) that specimens (aperture grids) irradiated by electrons can be imaged in magnified form not only in one but also in more than one stage by means of (magnetic) electron lenses.

(U = 50 kV). [8].  
a) one-stage image of the platinum grid in front of coil 1 by coil 1; M = 13 x  
b) one-stage image of the bronze grid in front of coil 2 by coil 2; M = 4.8 x  
c) two-stage image of the platinum grid in front of coil 1 by coil 2; M = 17.4 x together with the one-stage image of the bronze grid in front of coil 2 by coil 2; M = 4.8 x  
kk Cold cathode; Pt N Platinum grid; Sp 1 coil 1;  
Br N Bronze grid; Sp 2 coil 2; LS Fluorescent screen

Ernst Ruska, Nobel Lecture

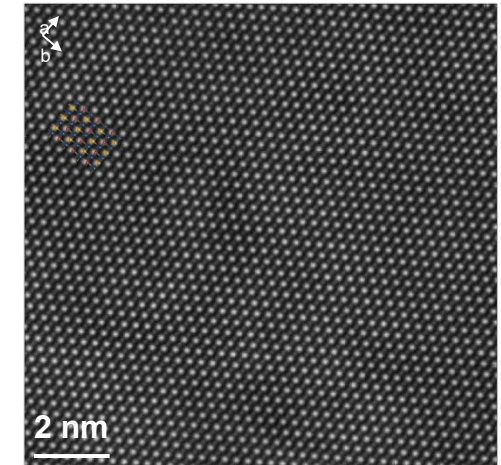
## Scanning tunneling microscope 1983



Binnig, Rohrer, et. al.  
Phys. Rev. Lett. 50, 120 (1983)

FIG. 2. Top view of the relief shown in Fig. 1 (the hill at the right is not included) clearly exhibiting the sixfold rotational symmetry of the maxima around the rhombohedron corners. Brightness is a measure of the altitude, but is not to scale. The crosses indicate adatom positions of the modified adatom model (see Fig. 3) or "milk-stool" positions (Ref. 5).

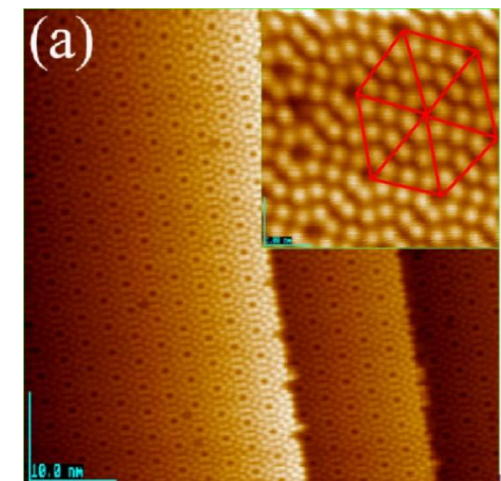
2025



Bulk (Volumen)

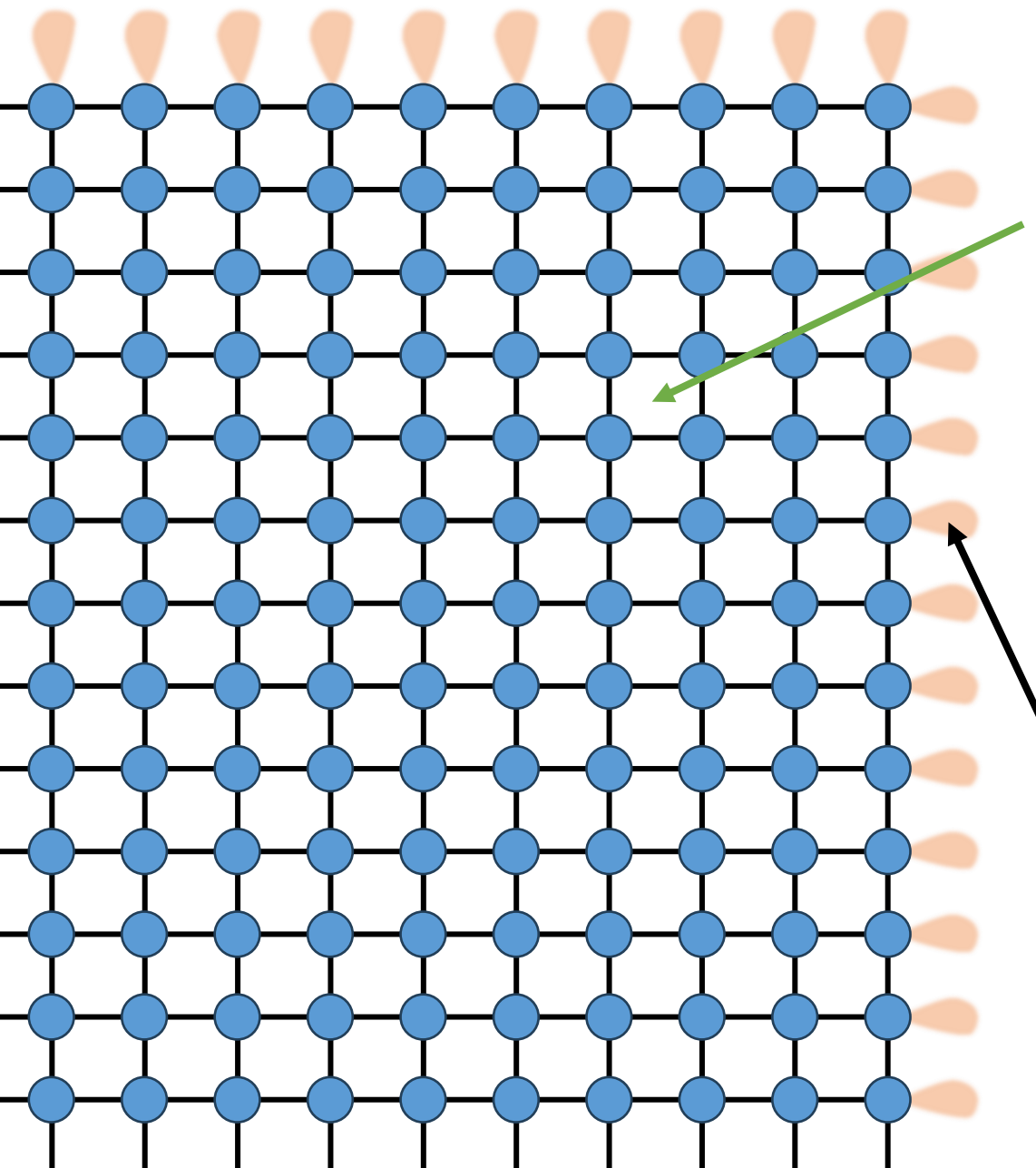
© Julian Klein, MIT

2014



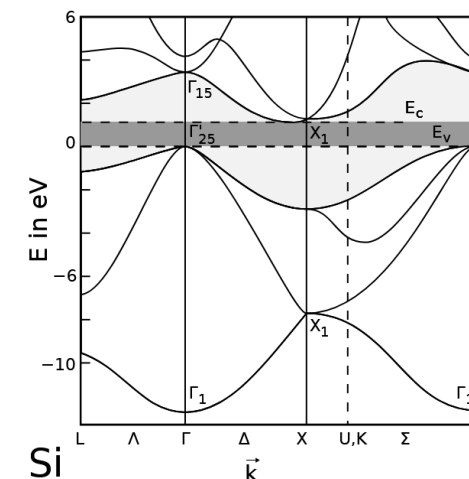
Surface (Oberfläche)

Nanoscale Research Letters 9, 377, (2014)



### Periodic crystal (main topic of this year):

- Periodicity  $\rightarrow$  Bloch Theorem  $\rightarrow$  simplified Schrödinger equation
- „Simple rules“ for allowed energy levels  $\rightarrow$  band structure



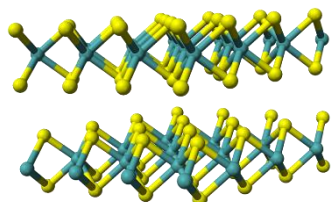
### What about the surface ???

- no periodicity, unsaturated bonds, reconstructions, ...



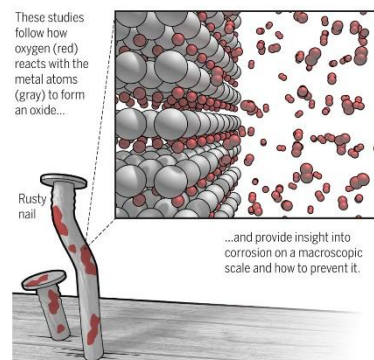
## Relevance of surfaces and interfaces:

- friction: microscopic interaction between surfaces
- chemical reactions:
  - surfaces or defects are active centers in catalysis
  - corrosion depends on surface properties
- electronics:
  - transport across interfaces (e.g. solar cell)
  - scattering from surfaces (Si-transistor)

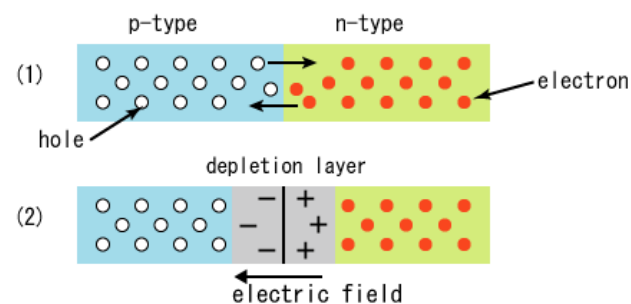


### Atomic oxidation visualized

Sun et al. used x-ray scattering and molecular dynamics simulations to follow oxidation of iron colloidal particles.



<https://www.science.org/doi/10.1126/science.aan0979>

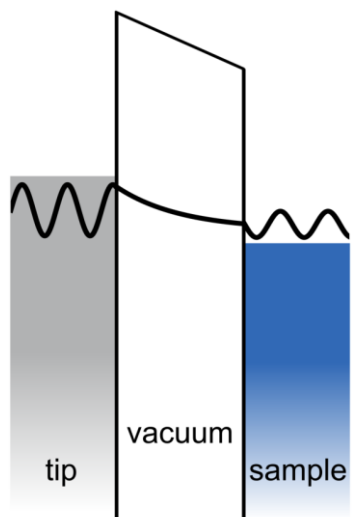


Wikipedia



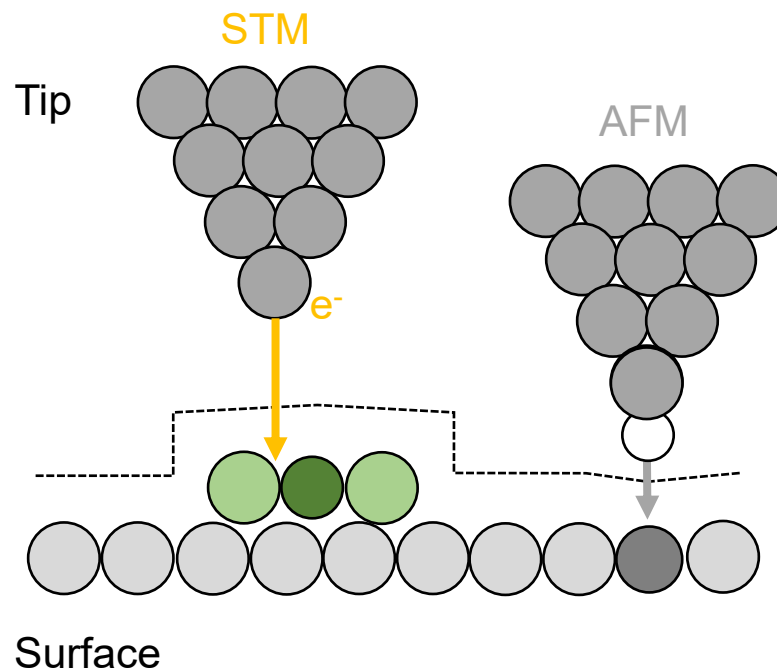
## Tunneling current

- due to an applied voltage
- due to local electronic structure



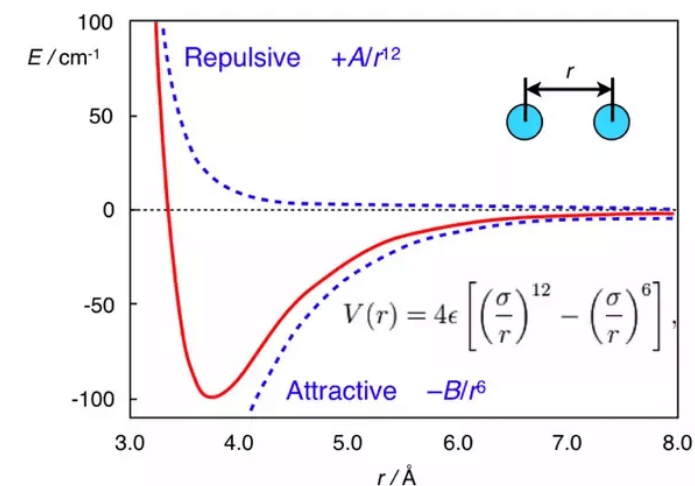
$$I \sim \exp(-d/z) \cdot \text{LDOS}$$

$$z_0 \sim \text{\AA} - \text{nm}$$

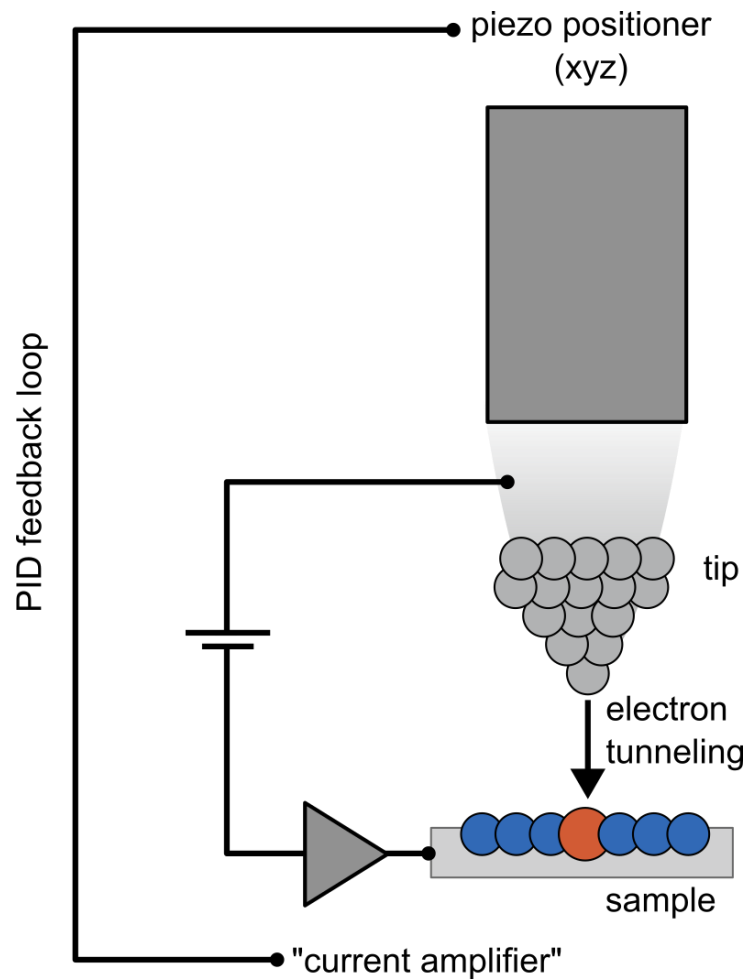


## Electrostatic force due

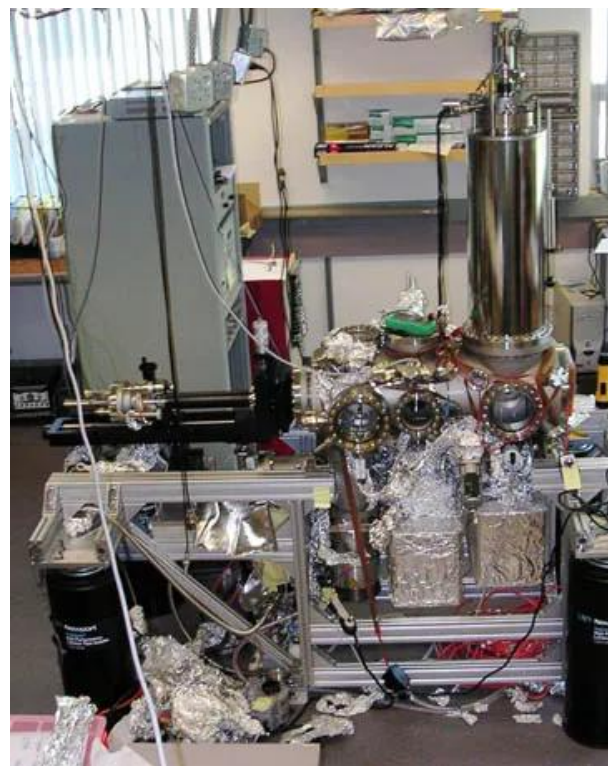
- to dipole interaction (vdW interaction)
- to Pauli repulsion (exchange interaction)







**UHV STM**



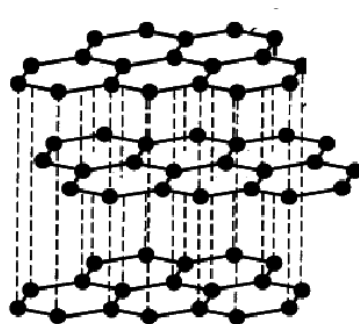
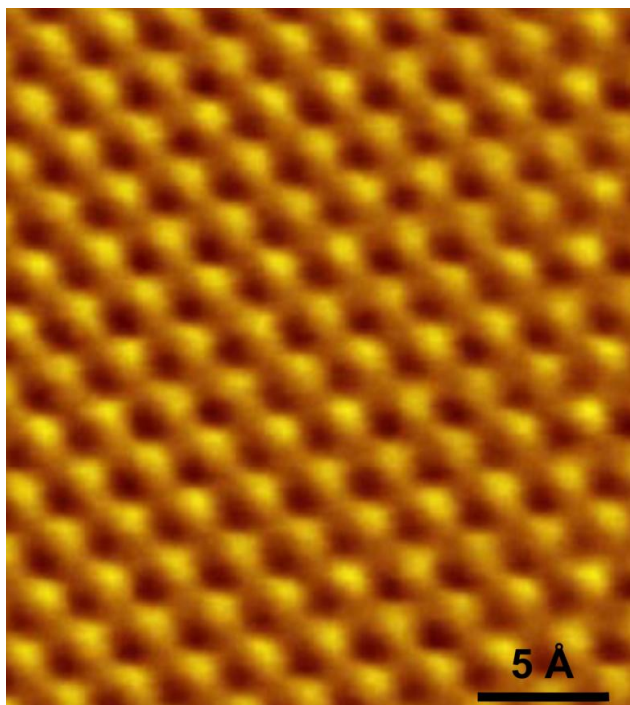
TUM Prof. Barth, Prof. Auwärter

**Ambient (or liquid) STM**

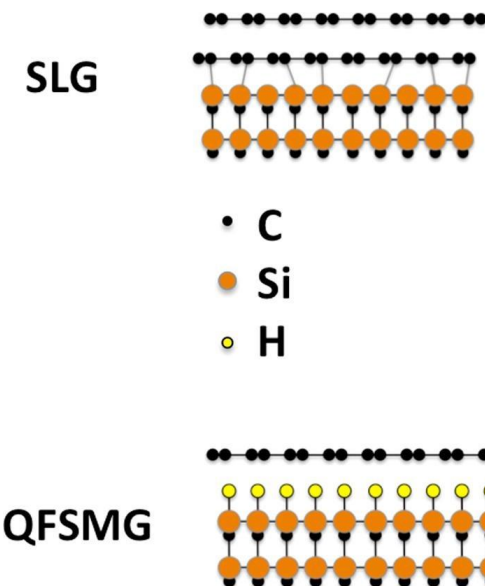
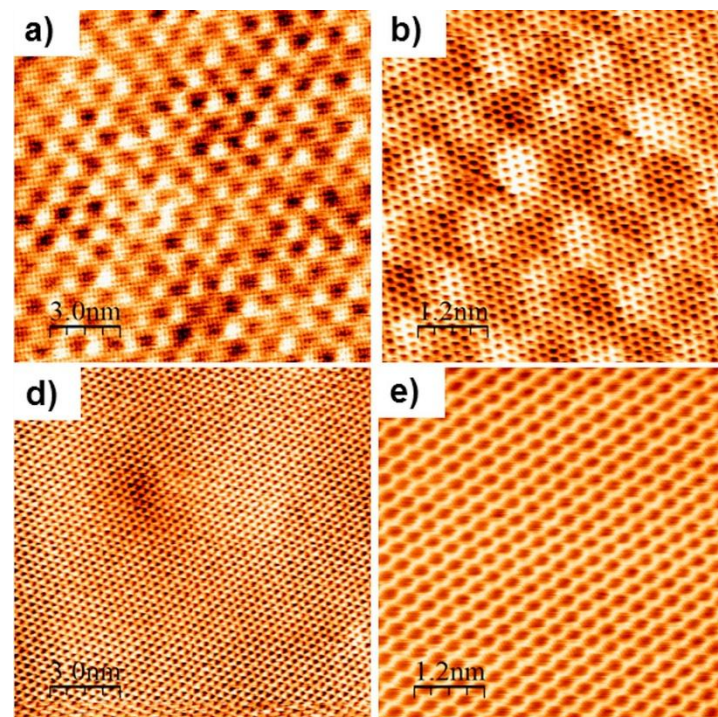


TUM Prof. Bandarenka  
TUM Prof. Eichhorn

## Example: HOPG, and graphene



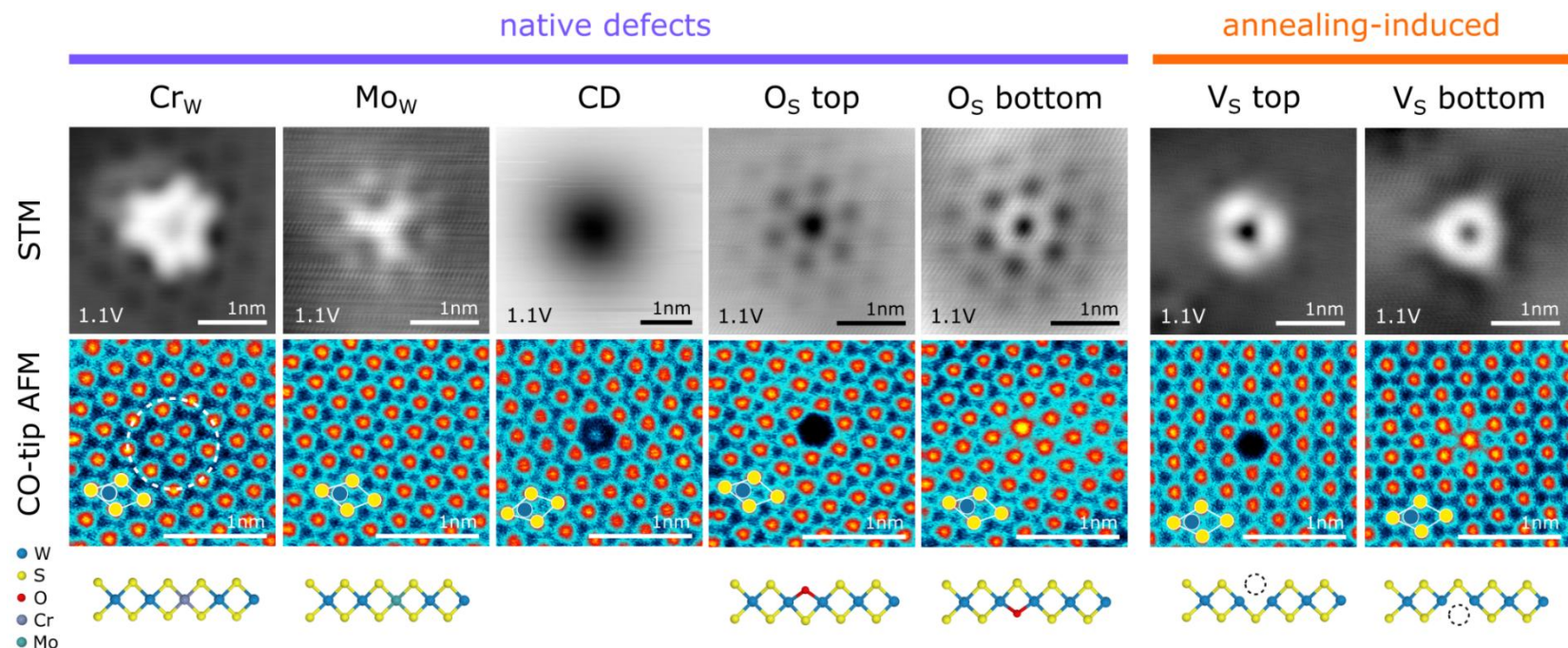
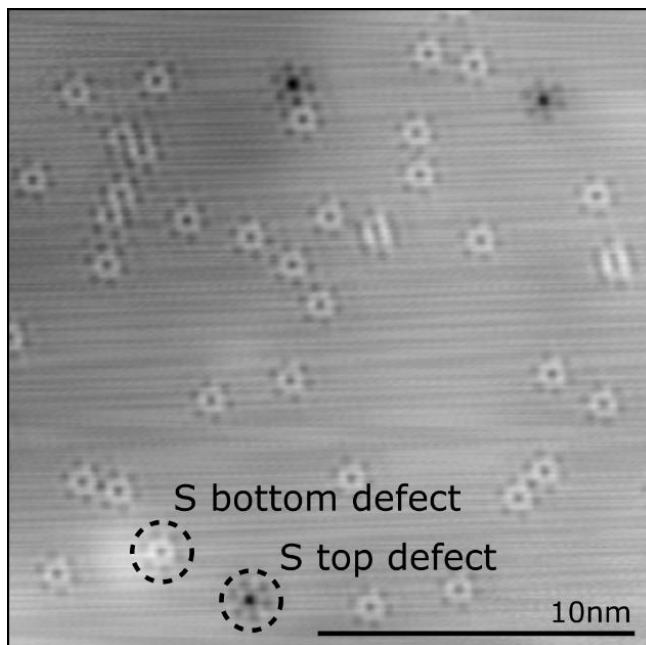
Solid State Commun. 149, 1151-1156 (2009)



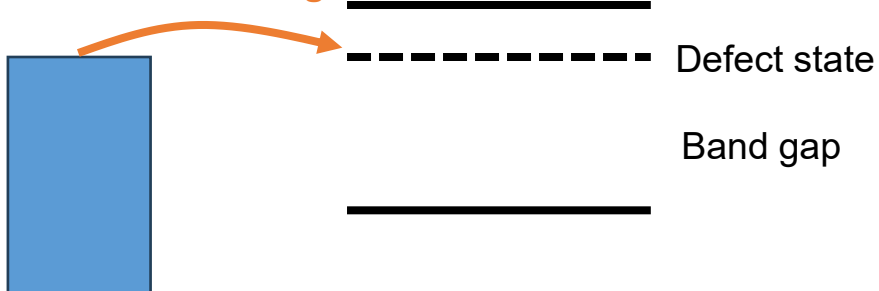
Applied Surface Science, 466, 2019, Pages 51-58

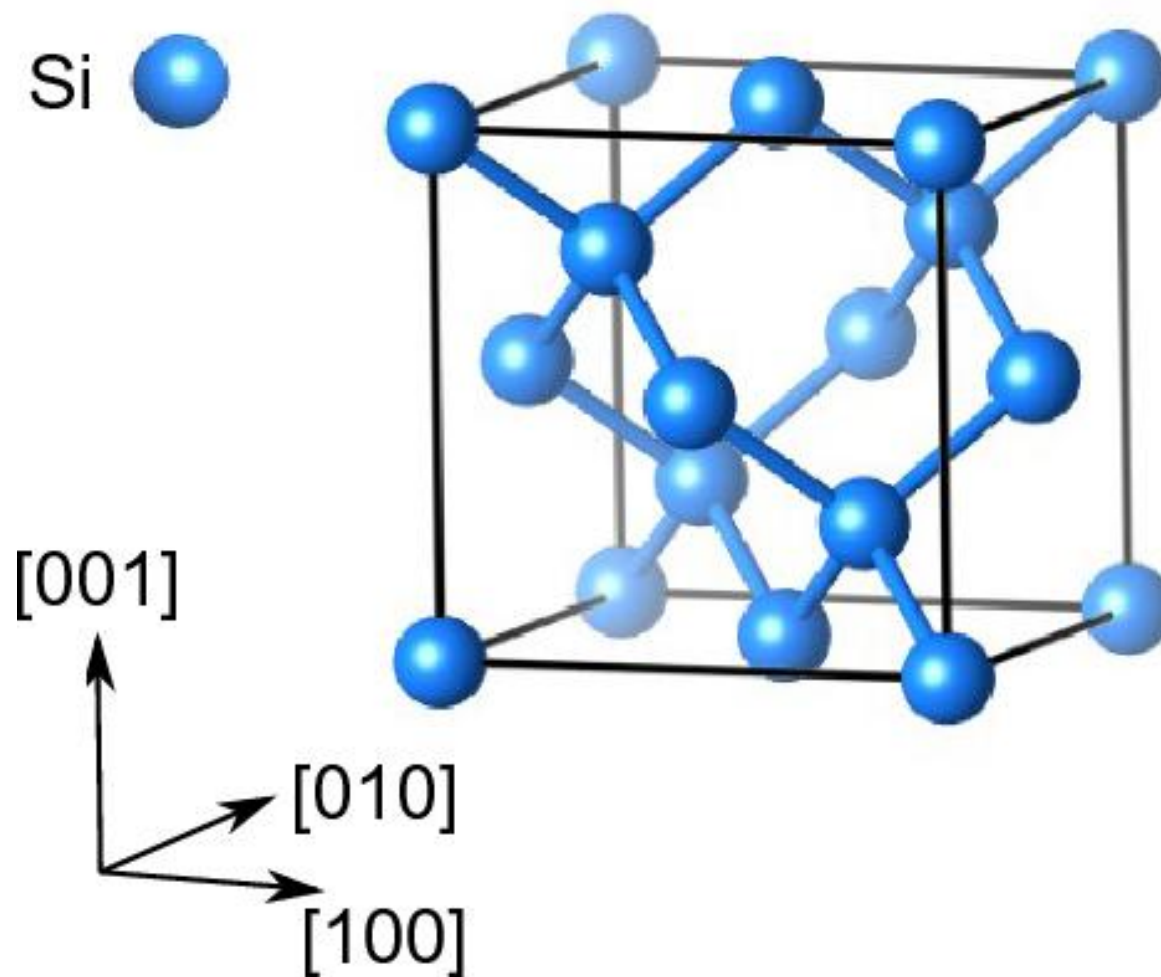


## Example: Defects in $WS_2$



resonant tunneling

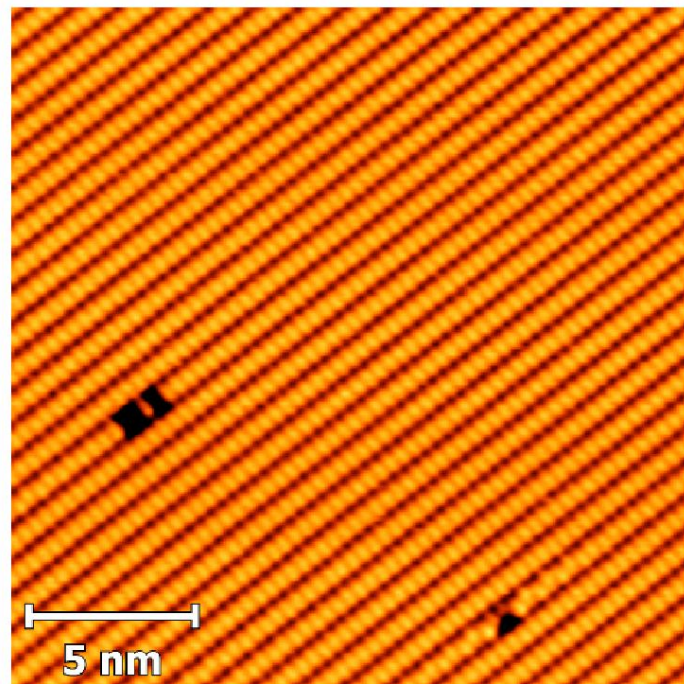
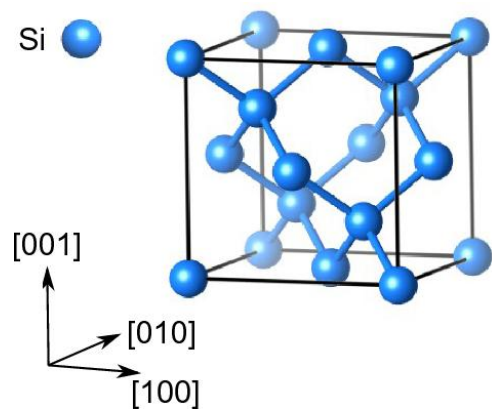






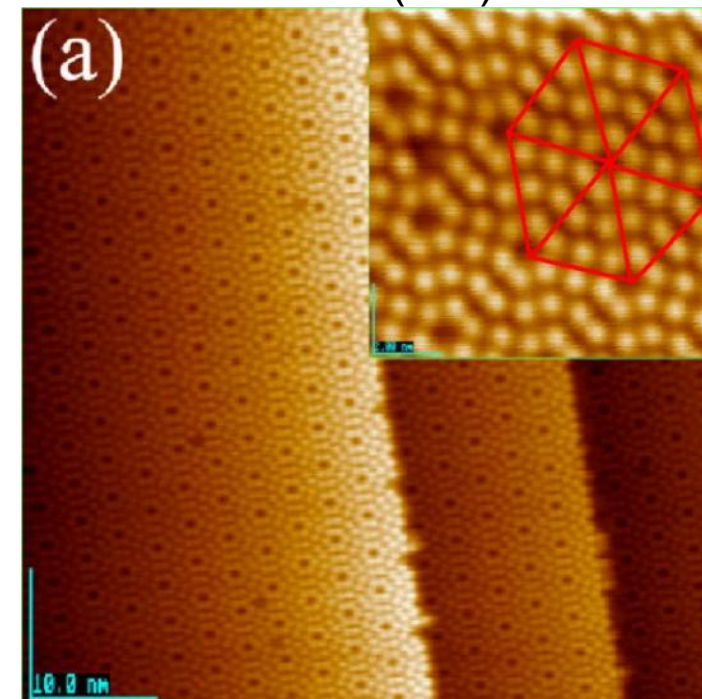
## Example: different surfaces of Si

Si (001)



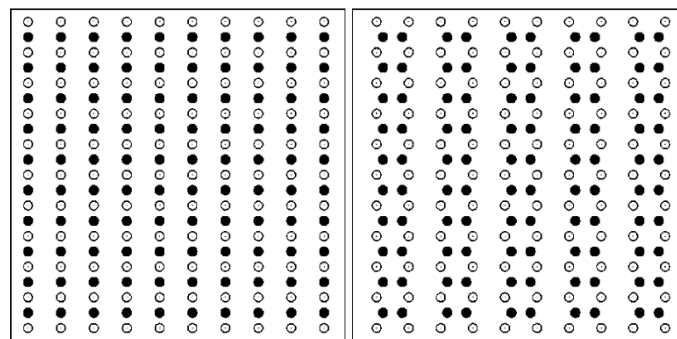
E. Crane et al 2018 J. Phys.: Conf. Ser. 1079 012010

Si (111)

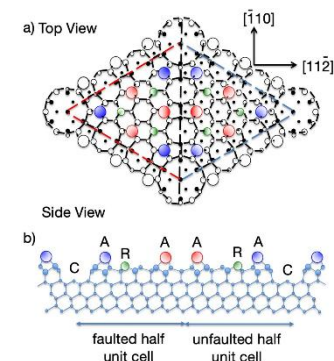


Nanoscale Research Letters 9, 377, (2014)

## Surface reconstruction

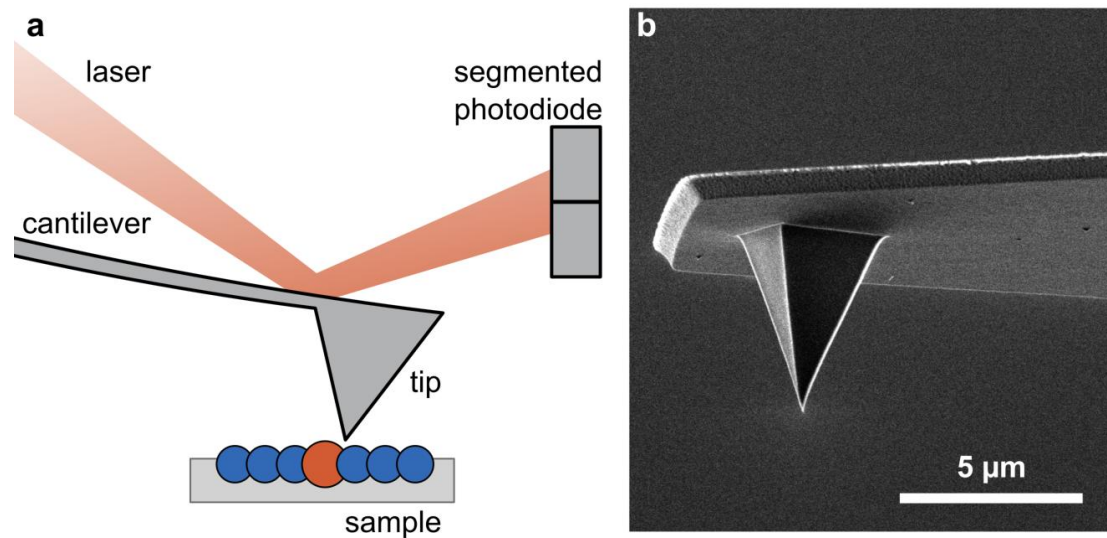


Phys. Rev. B 67, 035405 (2003)



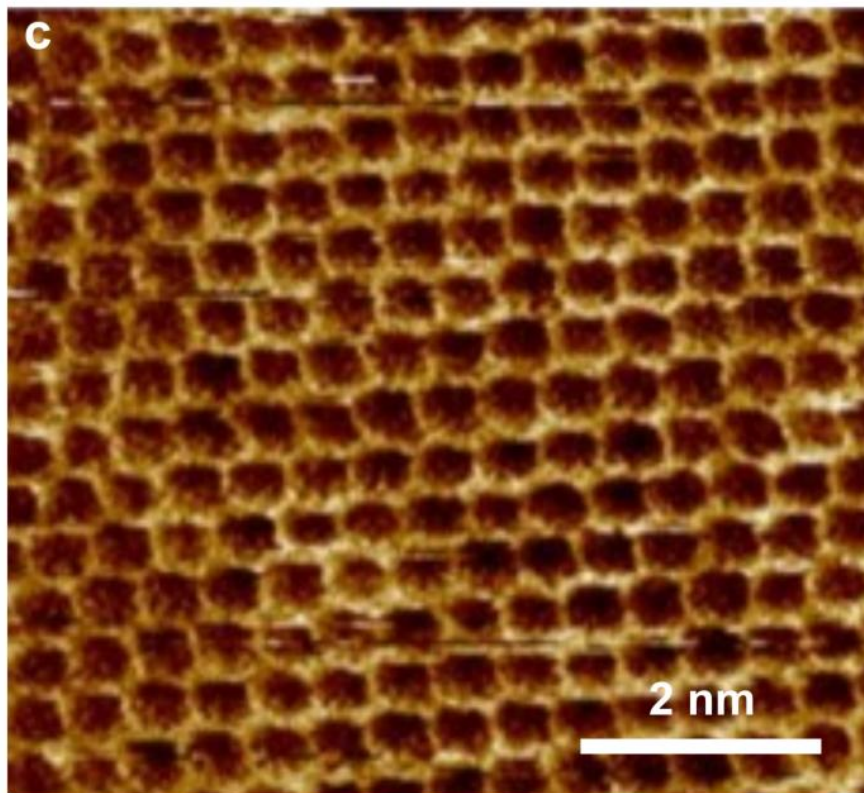
Scientific Reports 11, 15034 (2021)

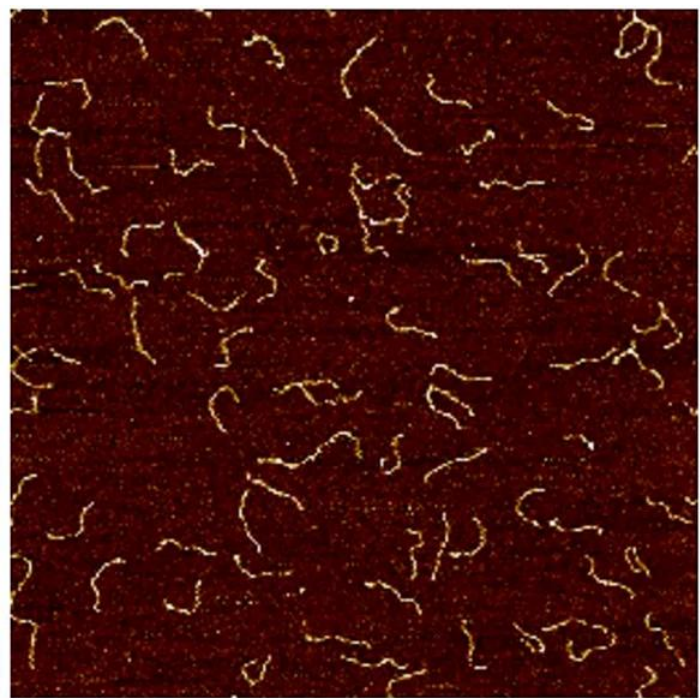
## Mechanical resonance





## *Example: Mica – „Glimmer“*



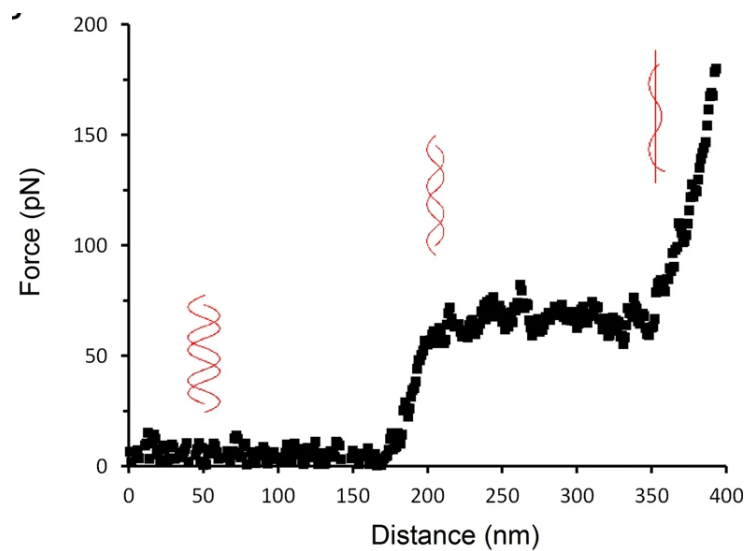
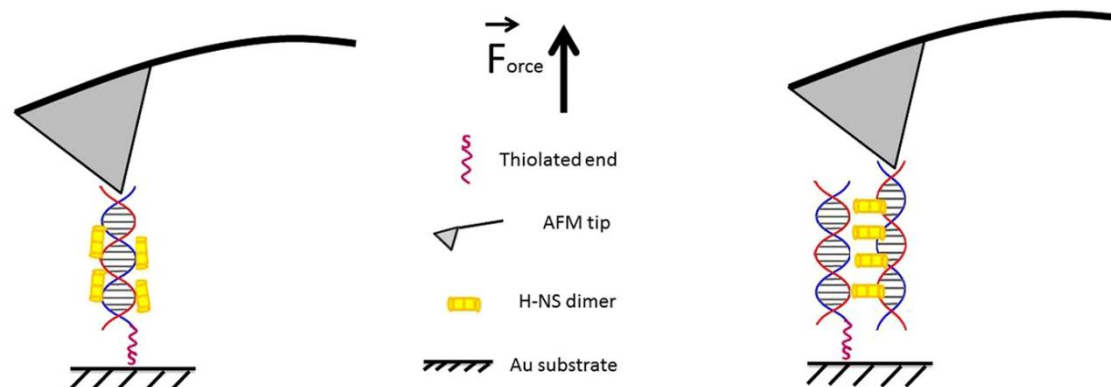


0.9 nm

-0.5 nm

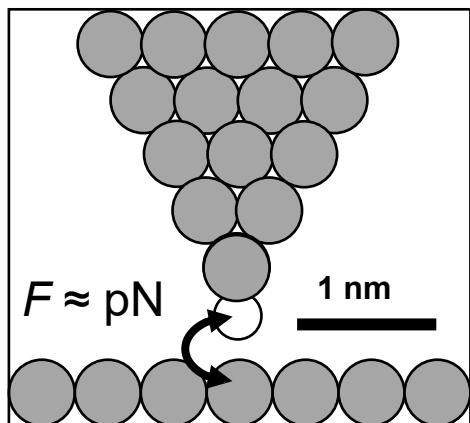
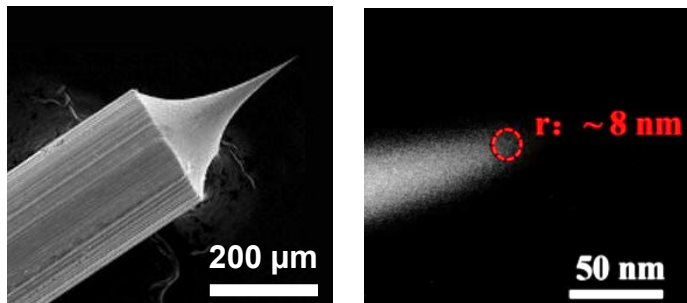
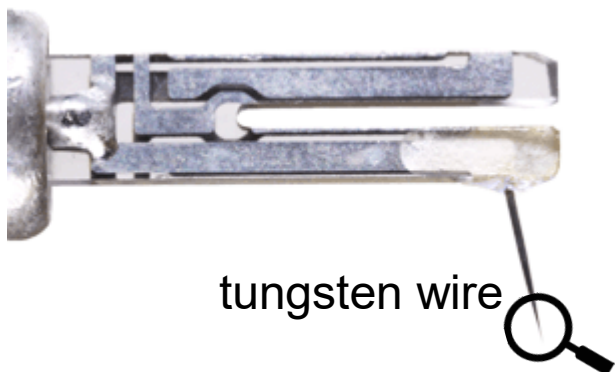
0.0 Height 2.0  $\mu\text{m}$

Scientific Reports 7, 15275 (2017).

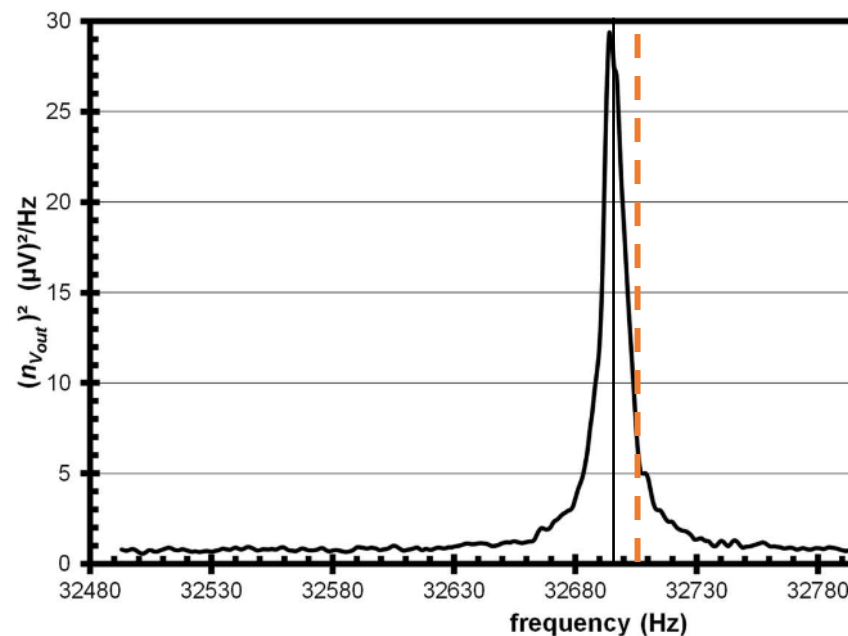


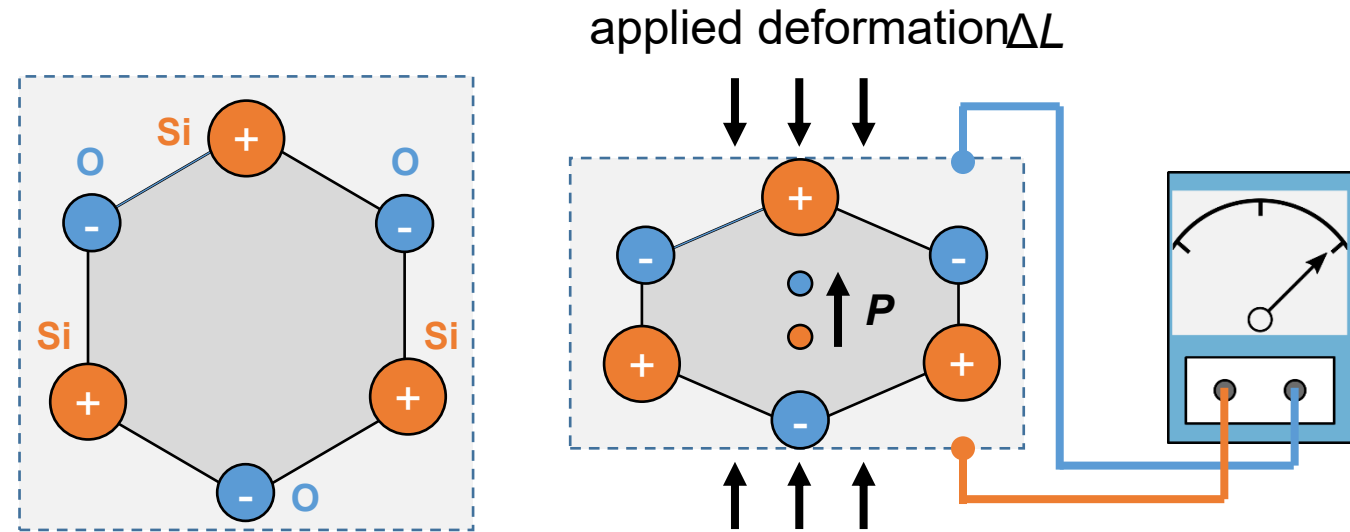
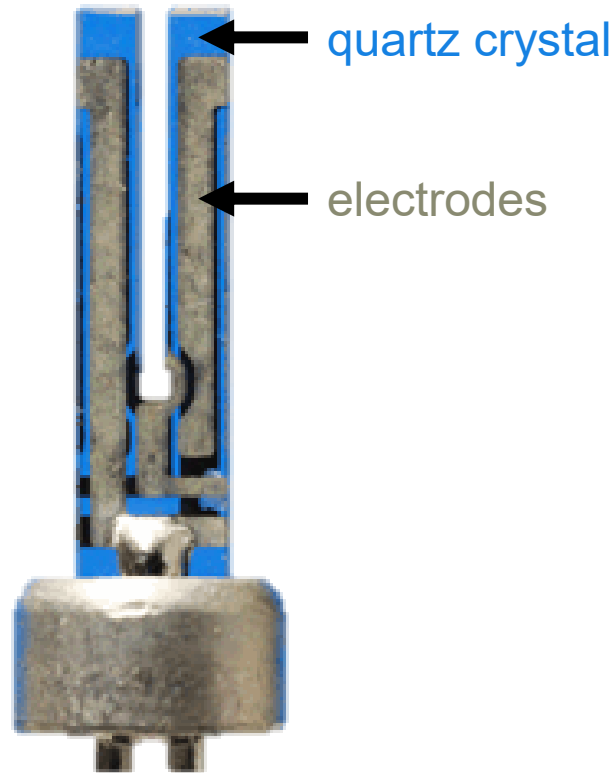


## Tuning fork AFM



- The interaction of the tip with a crystal surface results in by tiny electrostatic forces ( $F \approx \text{pN}$ )
- These forces appears as additional damping terms in the equation of motion.
- The additional damping shifts the resonance frequency of the oscillator





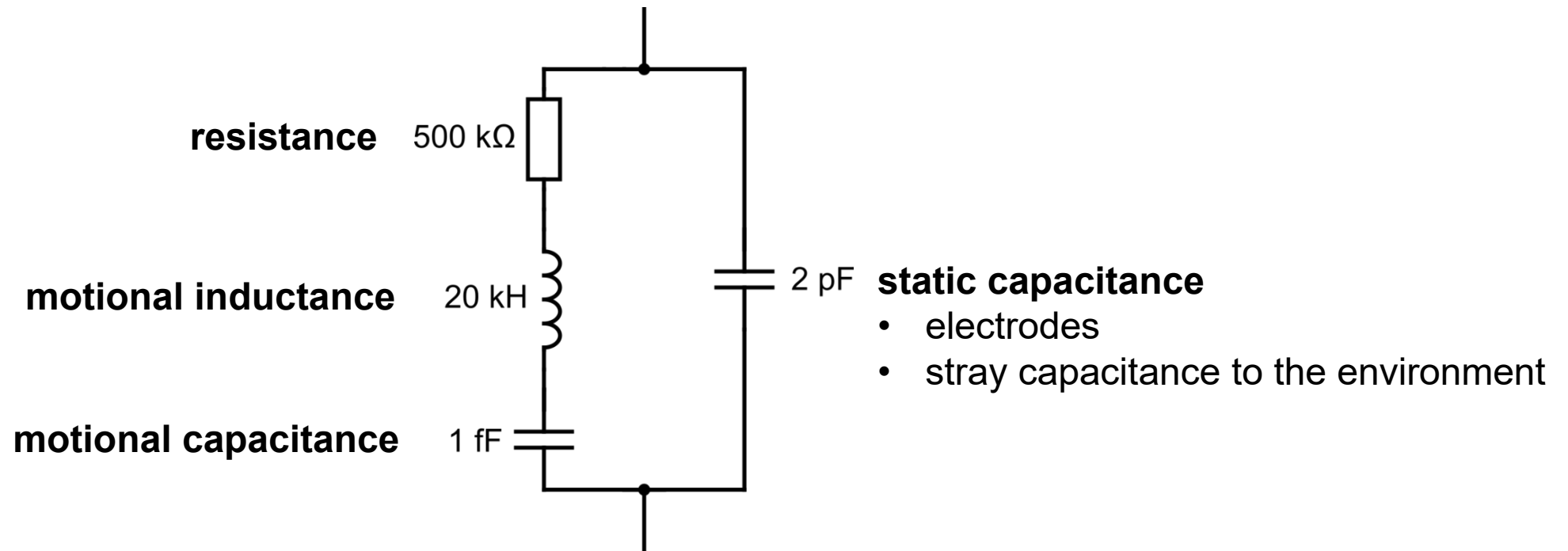
## Direct piezoelectric effect

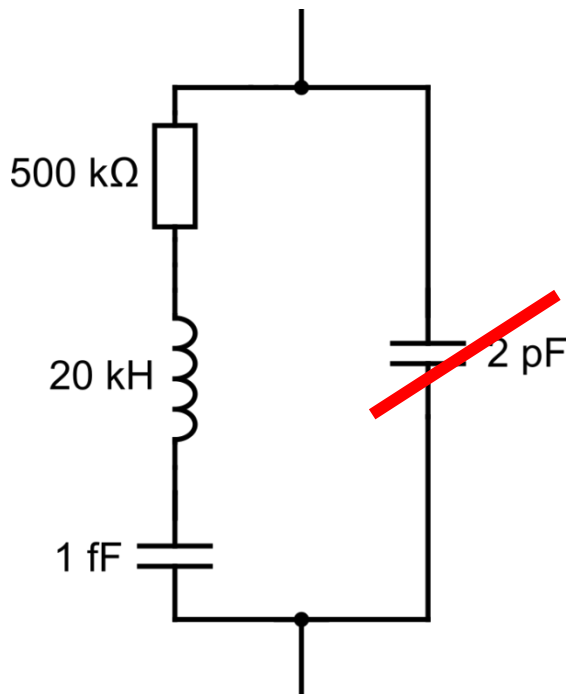
- Deformation of the crystal results in a polarisation  $P$ .
- The polarisation can be measured as a surface charge or voltage.
- This can be used in *sensors*.

## Inverse piezoelectric effect

- An applied field polarizes the crystal.
- The crystal responds with a deformation  $\Delta L$ .
- This can be used in *actuators*.

The quartz oscillator can be modelled by an equivalent electronic circuit





## Resonance frequency

$$\omega_{\text{res}}^2 = \frac{1}{LC} - \frac{R^2}{4L^2}$$

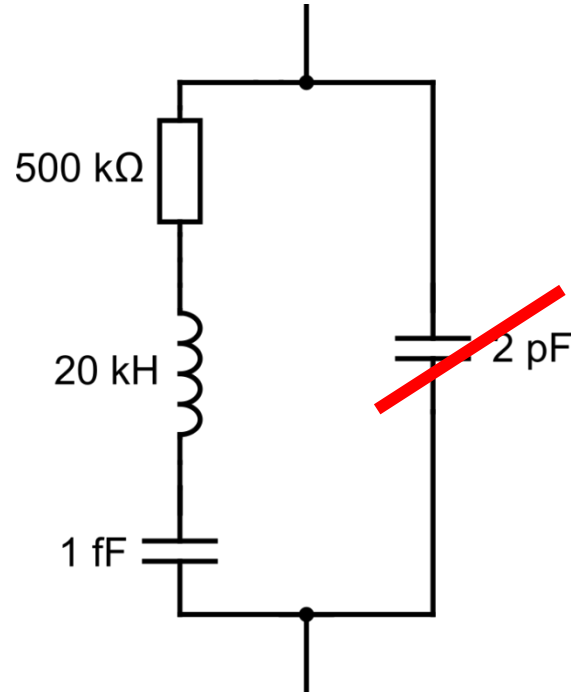
$$\omega_{\text{res}}^2 = \frac{1}{2 \cdot 10^4 \text{ H} \cdot 10^{-15} \text{ F}} - \frac{1}{4} \left( \frac{5 \cdot 10^5 \Omega}{2 \cdot 10^4 \text{ H}} \right)^2$$

$$\omega_{\text{res}}^2 = \frac{1}{2} 10^{11} \text{ s}^{-2} - \left( \frac{5}{4} \right)^2 \cdot 100 \text{ s}^{-2}$$

$$\omega_{\text{res}} \approx \omega_0 = 223\,000 \text{ s}^{-1}$$

$$f_0 \approx 35.5 \text{ kHz}$$



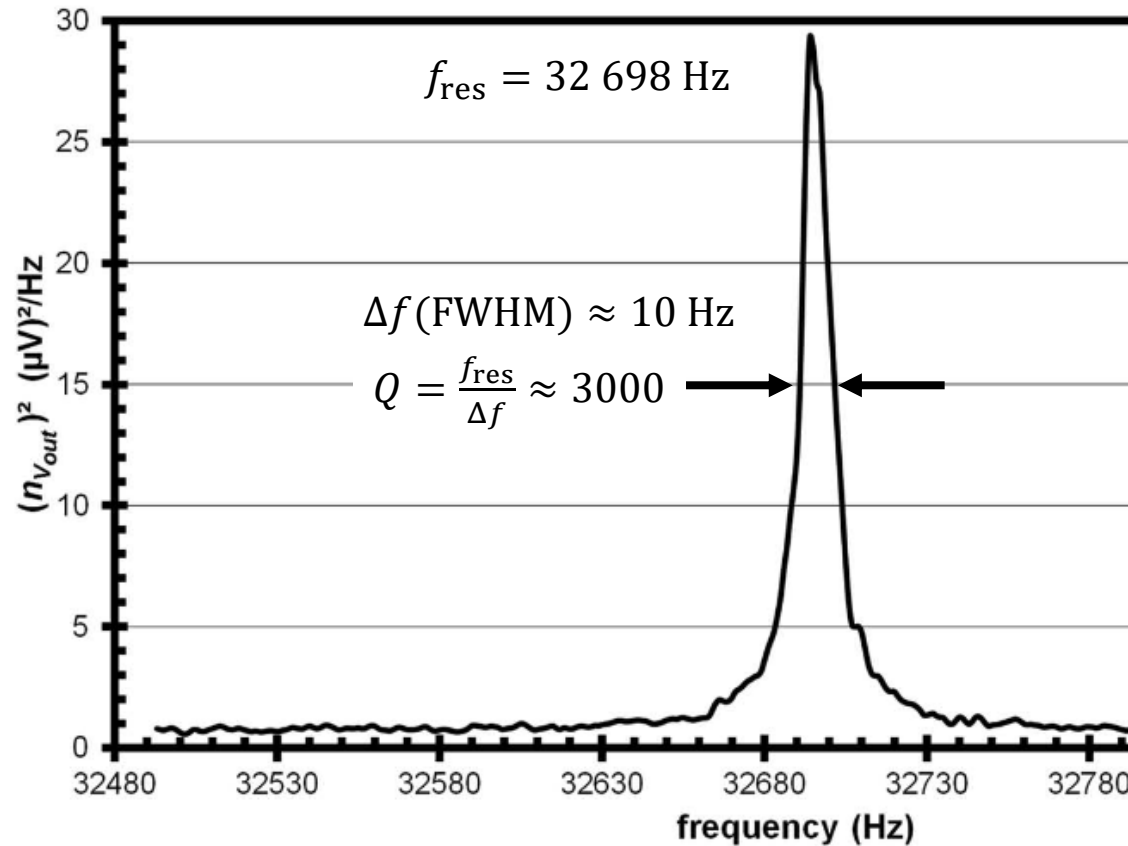


**Q-factor**

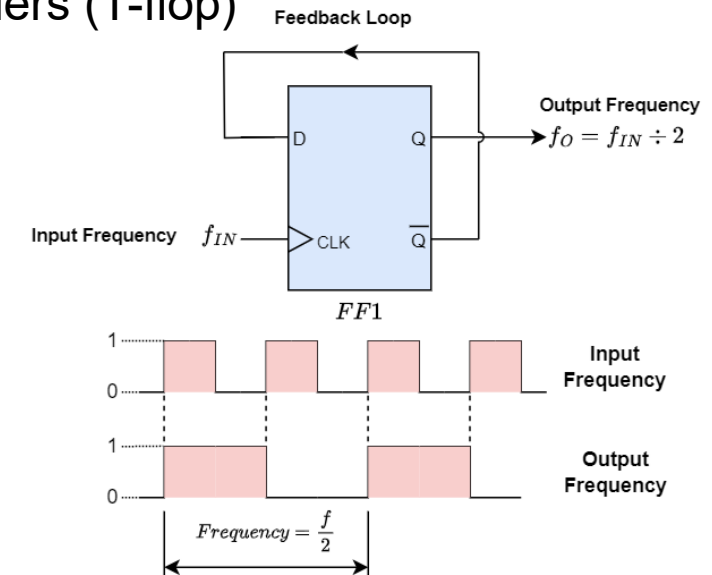
$$Q = R \sqrt{\frac{L}{C}} = \frac{\omega_0 L}{R} = \frac{1}{\omega_0 RC}$$

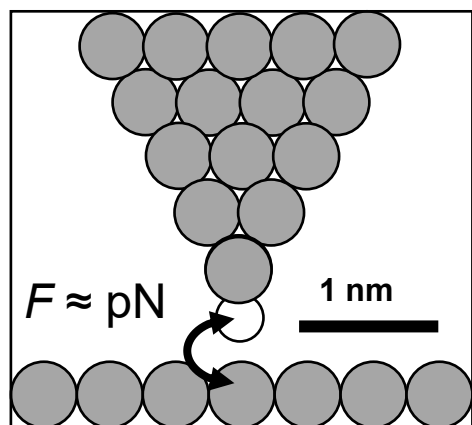
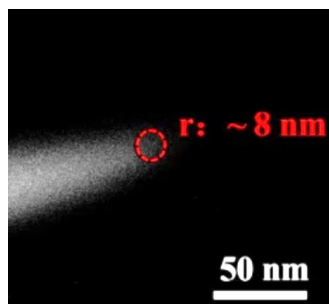
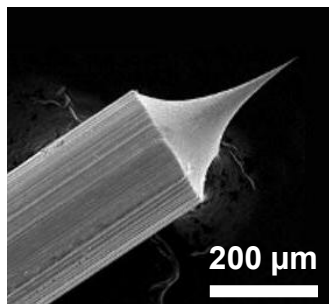
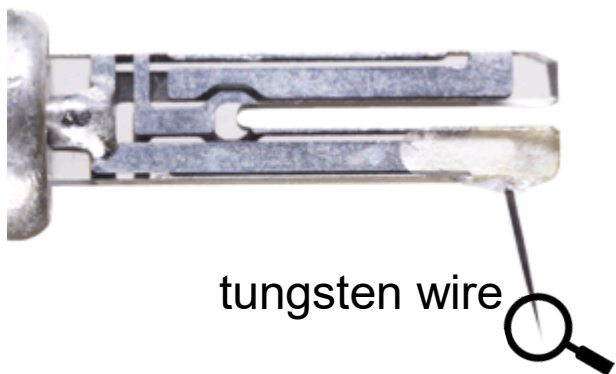
$$Q = \frac{1}{2\pi \cdot 35.5 \text{ kHz} \cdot 500 \text{ k}\Omega \cdot 1 \text{ fF}}$$

$$Q = \frac{1}{10\pi \cdot 35.5} 10^7 \approx 9000$$

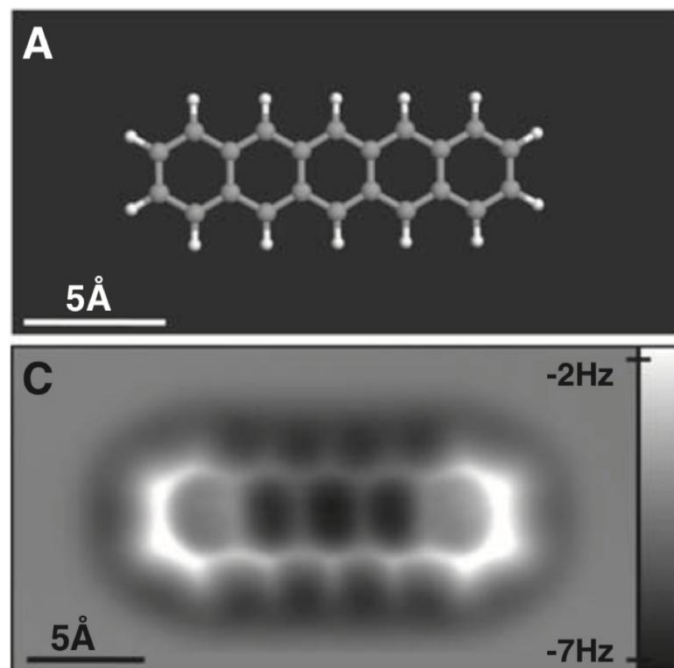


- $2^{15} = 32\,768\text{ Hz}$
- Used as frequency reference in wrist watches
- 1 Hz is derived from  $f_0$  by a cascade of digital frequency dividers (T-flop)

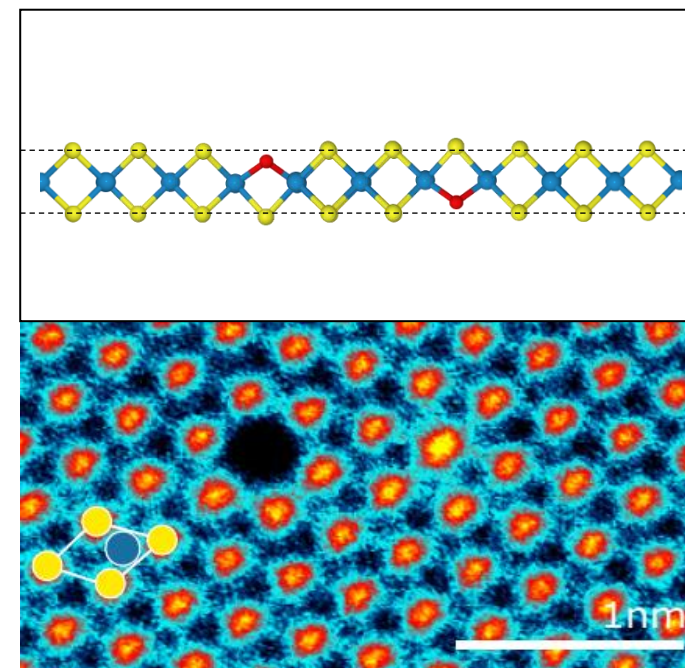




Bonds of an organic molecule



Atomic defects in a 2D crystal





# **Experimental methods in solid-state physics**

## **03 Real space structure – TEM**

Dr. Christoph Kastl  
Walter Schottky Institut and Physics Department  
Zentrum für Nanotechnologie  
und Nanomaterialien



# Nobel Prize in Physics 1986



Photo: Marianne Kischke, Fritz Haber Institut der Max Planck Gesellschaft. Nobel Foundation archive.

Ernst Ruska

Prize share: 1/2

Fritz-Haber-Institut (Berlin)



Photo from the Nobel Foundation archive.

Gerd Binnig

Prize share: 1/4



Photo from the Nobel Foundation archive.

Heinrich Rohrer

Prize share: 1/4

IBM (Rüschlikon)

The Nobel Prize in Physics 1986 was divided, one half awarded to Ernst Ruska "for his fundamental work in electron optics, and for the design of the first electron microscope", the other half jointly to Gerd Binnig and Heinrich Rohrer "for their design of the scanning tunneling microscope"

To cite this section

MLA style: The Nobel Prize in Physics 1986. NobelPrize.org. Nobel Prize Outreach 2025. Mon. 27 Oct 2025. <<https://www.nobelprize.org/prizes/physics/1986/summary/>>

## Transmission electron microscope 1931

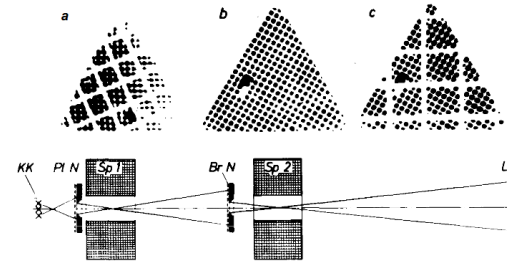


Fig. 3: First experimental proof (7 April 1931) that specimens (aperture grids) irradiated by electrons can be imaged in magnified form not only in one but also in more than one stage by means of (magnetic) electron lenses. (U = 50 kV). [8].

a) one-stage image of the platinum grid in front of coil 1 by coil 1; M = 13 x  
b) one-stage image of the bronze grid in front of coil 2 by coil 2; M = 4.8 x  
c) two-stage image of the platinum grid in front of coil 1 by coil 2; M = 17.4 x together with the one-stage image of the bronze grid in front of coil 2 by coil 2; M = 4.8 x  
kk Cold cathode; Pt N Platinum grid; Sp 1 coil 1;  
Br N Bronze grid; Sp 2 coil 2; LS Fluorescent screen

Ernst Ruska, Nobel Lecture

## Scanning tunneling microscope 1983

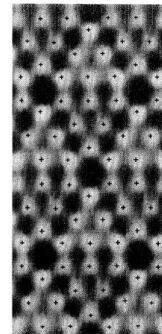
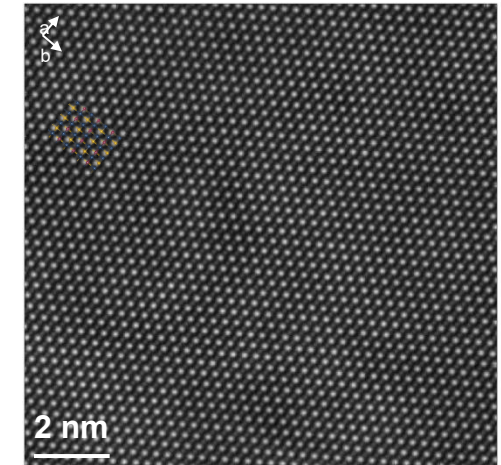


FIG. 2. Top view of the relief shown in Fig. 1 (the hill at the right is not included) clearly exhibiting the sixfold rotational symmetry of the maxima around the rhombohedron corners. Brightness is a measure of the altitude, but is not to scale. The crosses indicate adatom positions of the modified adatom model (see Fig. 3) or "milk-stool" positions (Ref. 5).

Binnig, Rohrer, et. al.  
Phys. Rev. Lett. 50, 120 (1983)

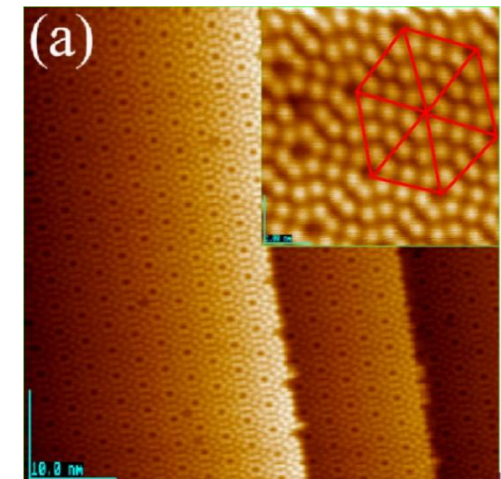
2025



Bulk (Volumen)

© Julian Klein, MIT

2014



Surface (Oberfläche)

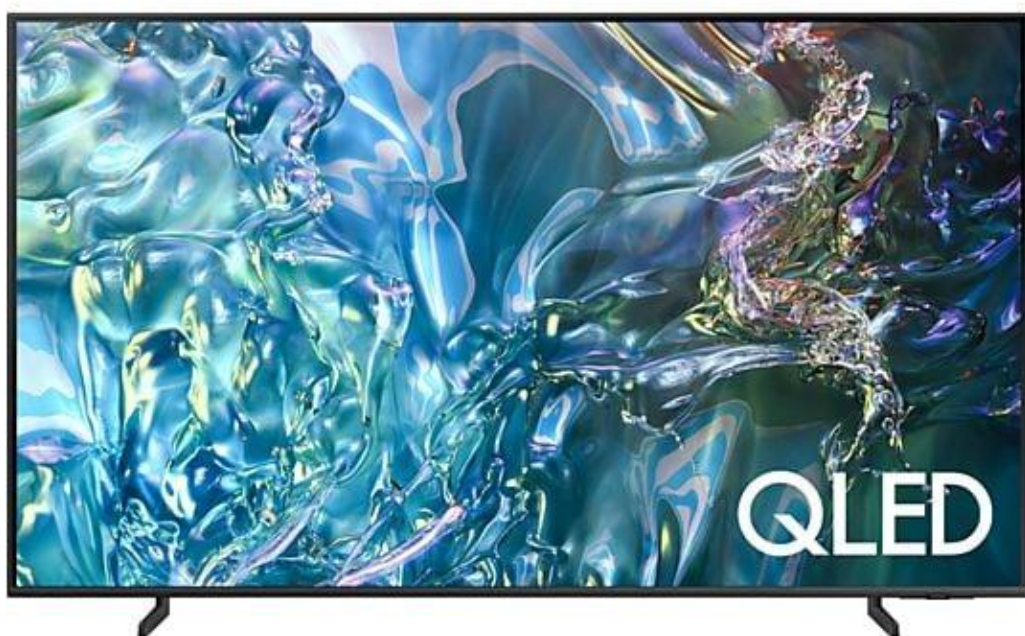
Nanoscale Research Letters 9, 377, (2014)



# *What do Samsung TVs have in common with the Nobel prize in chemistry 2023*



## Nobel Prize in Chemistry 2023



© Nobel Prize Outreach. Photo: Clément Morin

**Mounqi G. Bawendi**

Prize share: 1/3



© Nobel Prize Outreach. Photo: Clément Morin

**Louis E. Brus**

Prize share: 1/3

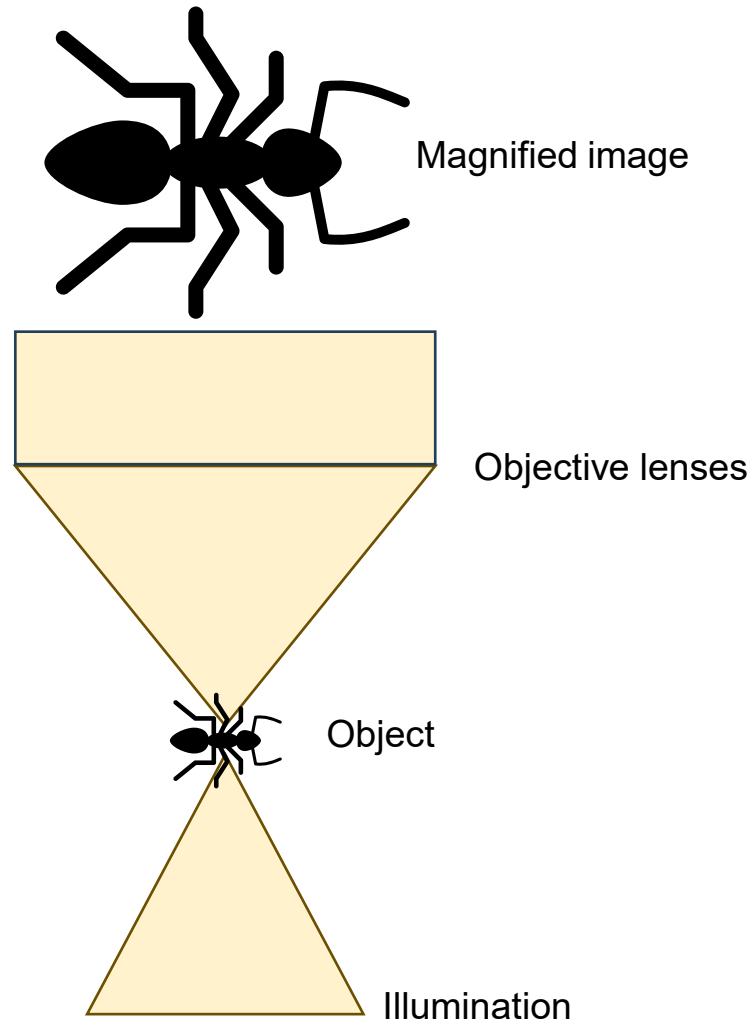


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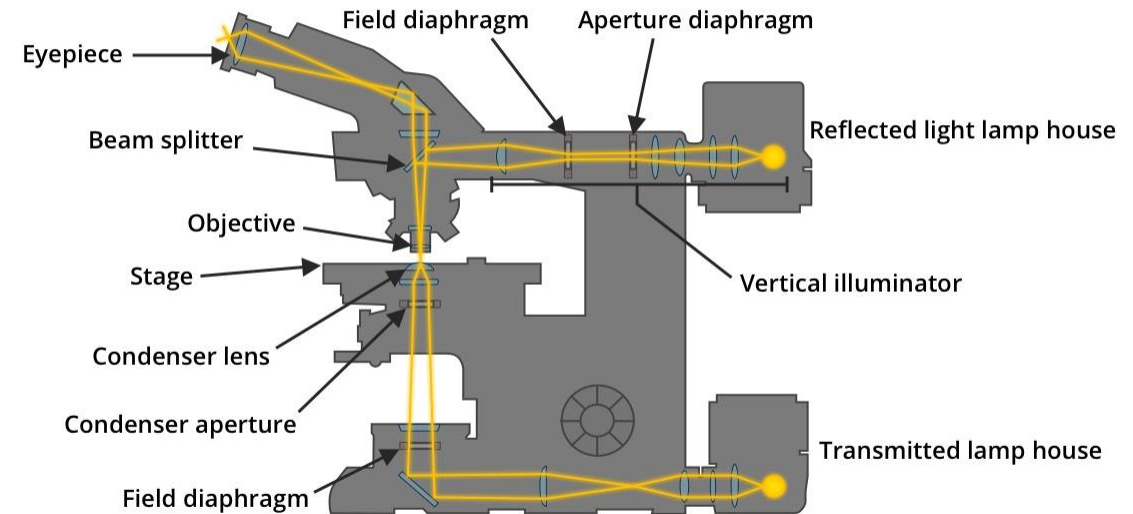
**Aleksey Yekimov**

Prize share: 1/3





## Reflected/Transmitted Light Microscope



### Magnification:

- $\text{Mag1} \times \text{Mag2} \times \text{Mag3} \dots$
- (arbitrarily) large

### Resolution:

- $\lambda / (2 \text{ NA}) \rightarrow$  diffraction limit in the far field
- 532 nm (green),  $\text{NA} \sim 1 \rightarrow 266 \text{ nm}$

## Contrast-Enhancing Techniques in Optical Microscopy

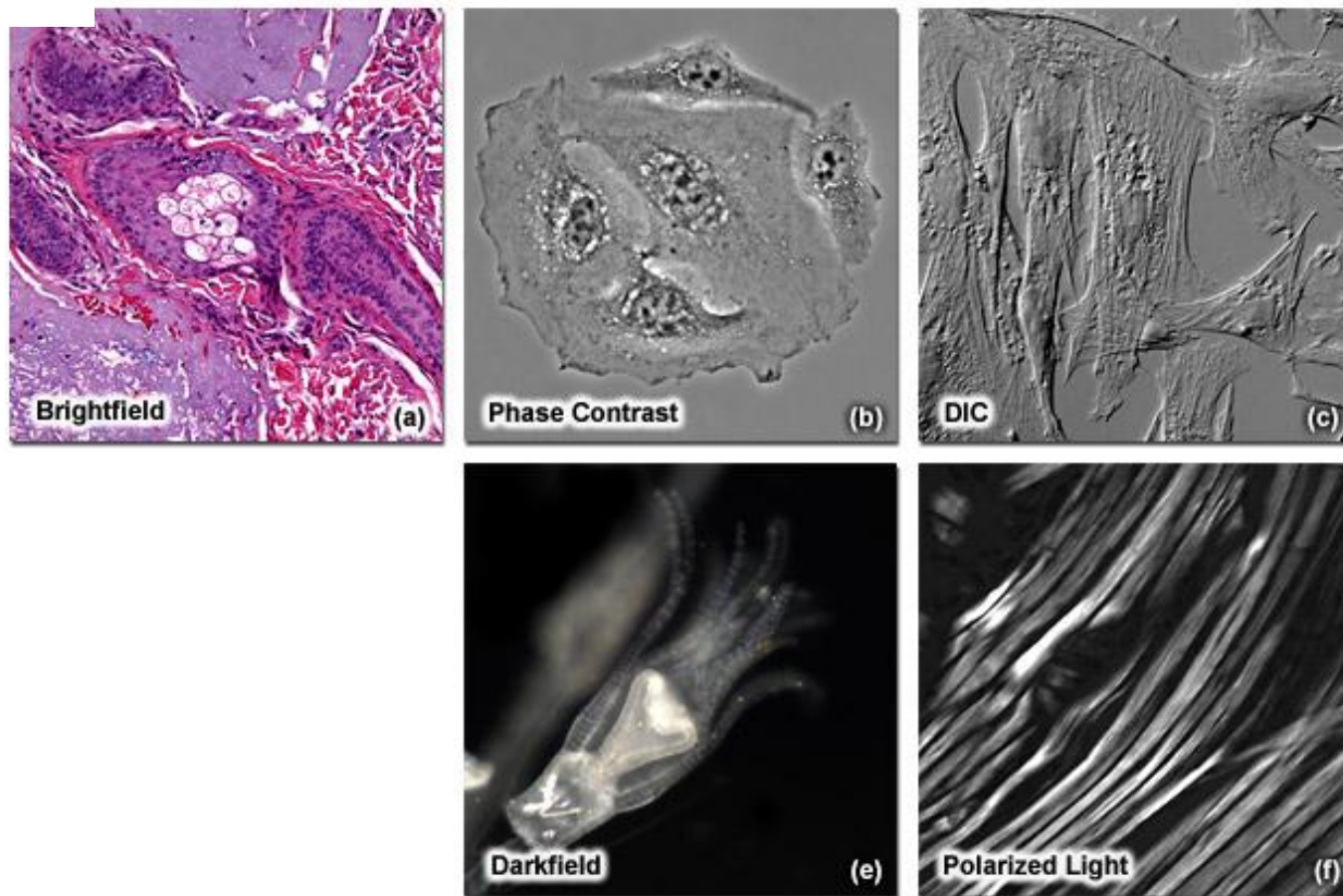
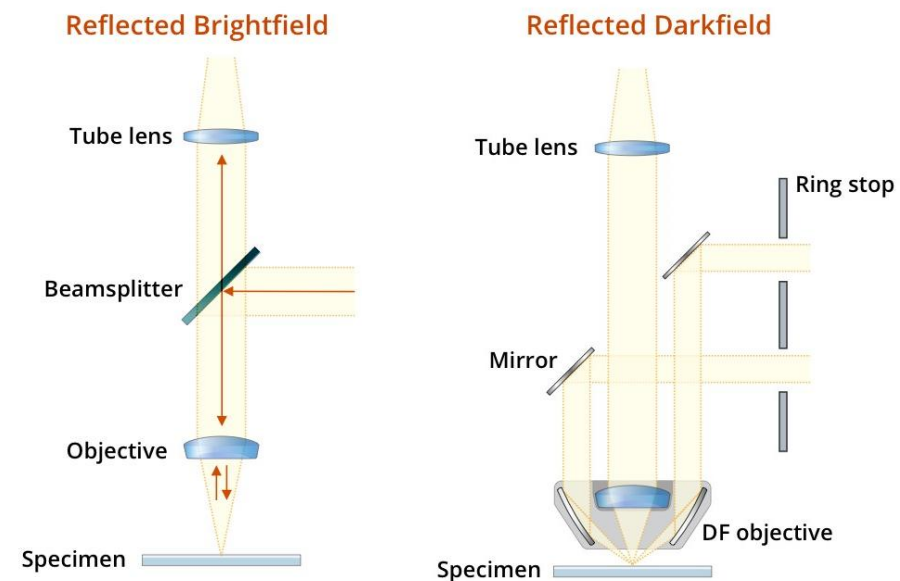


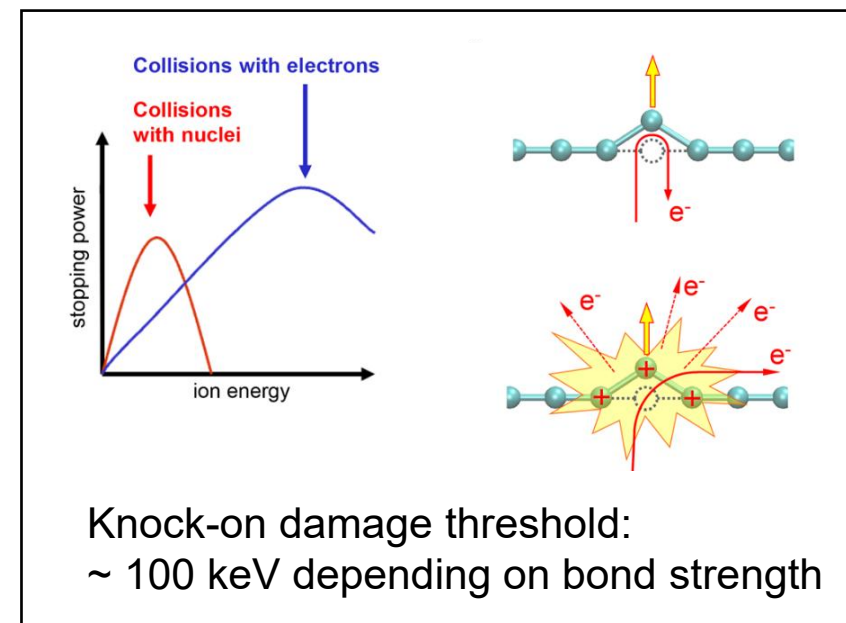
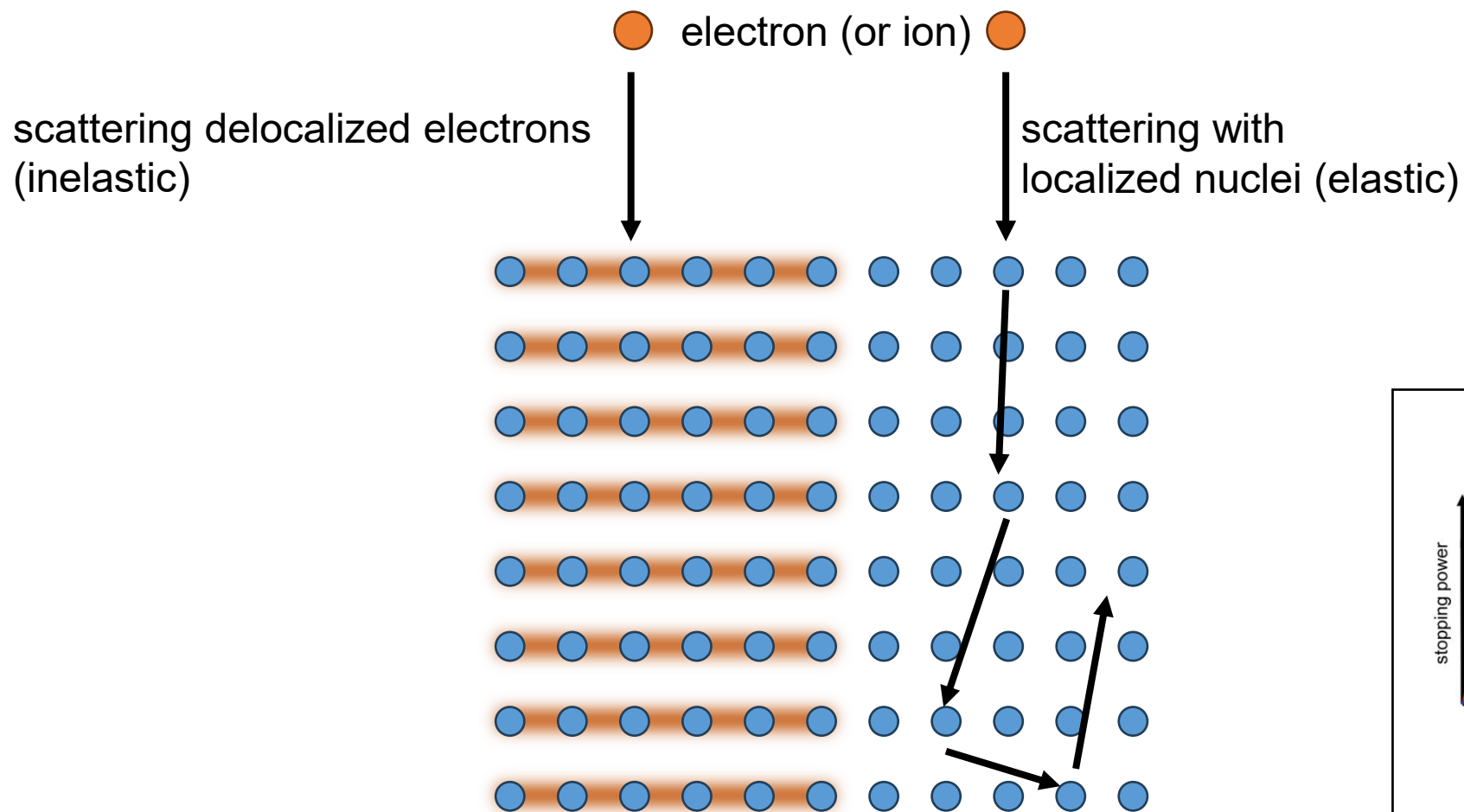
Figure 1

Source: Zeiss.

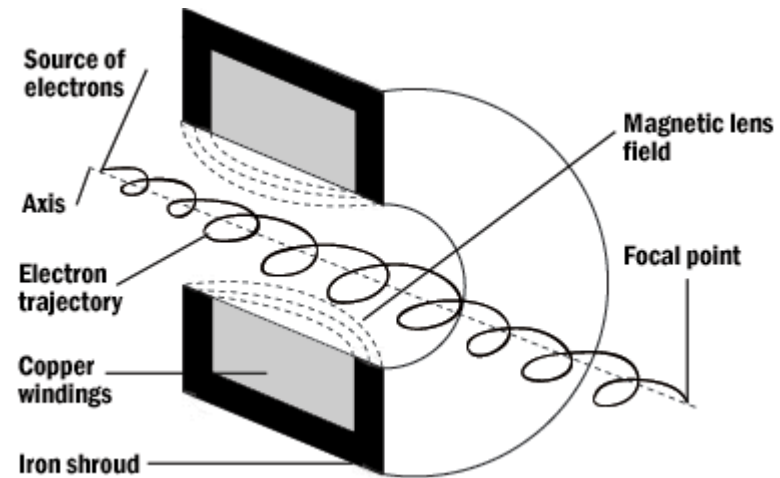


Source: myscope.training.

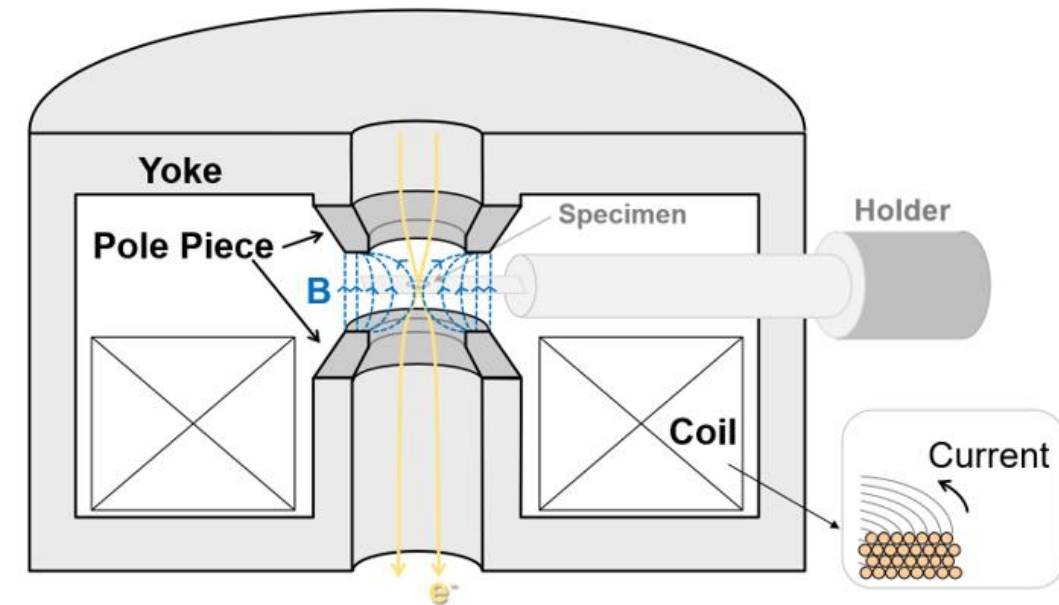




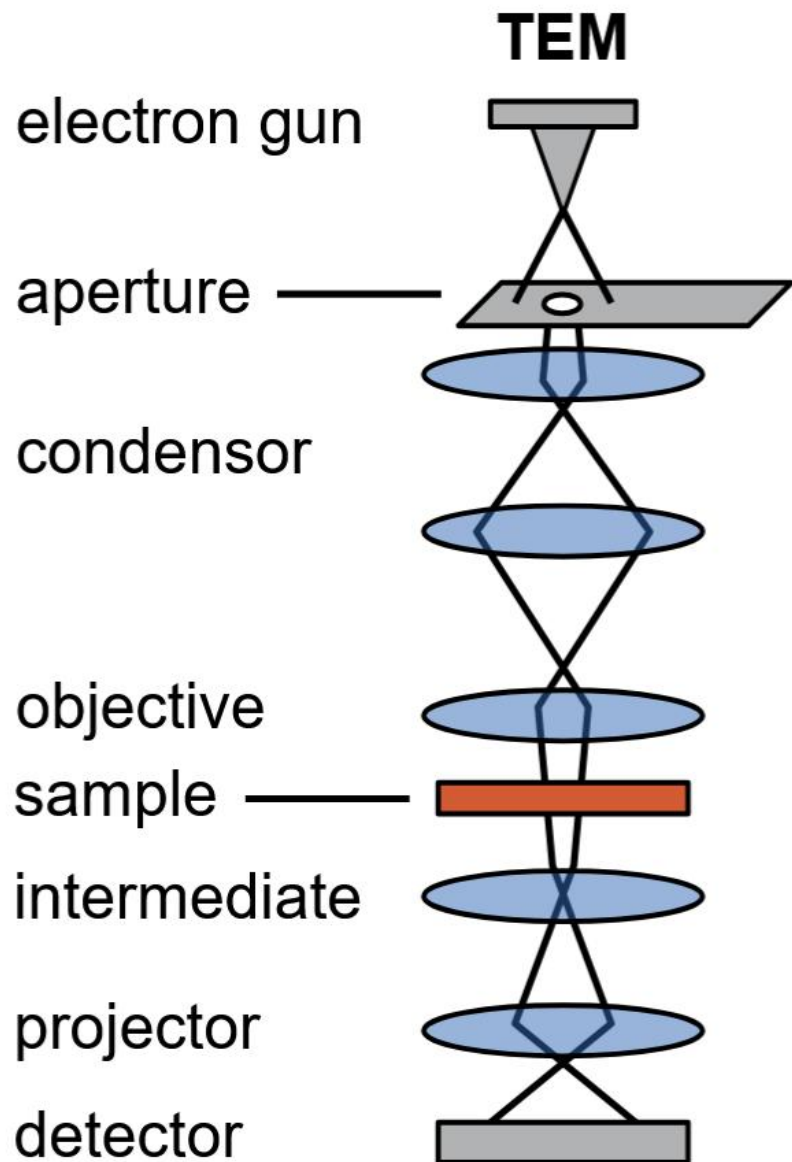
## Magnetic lens



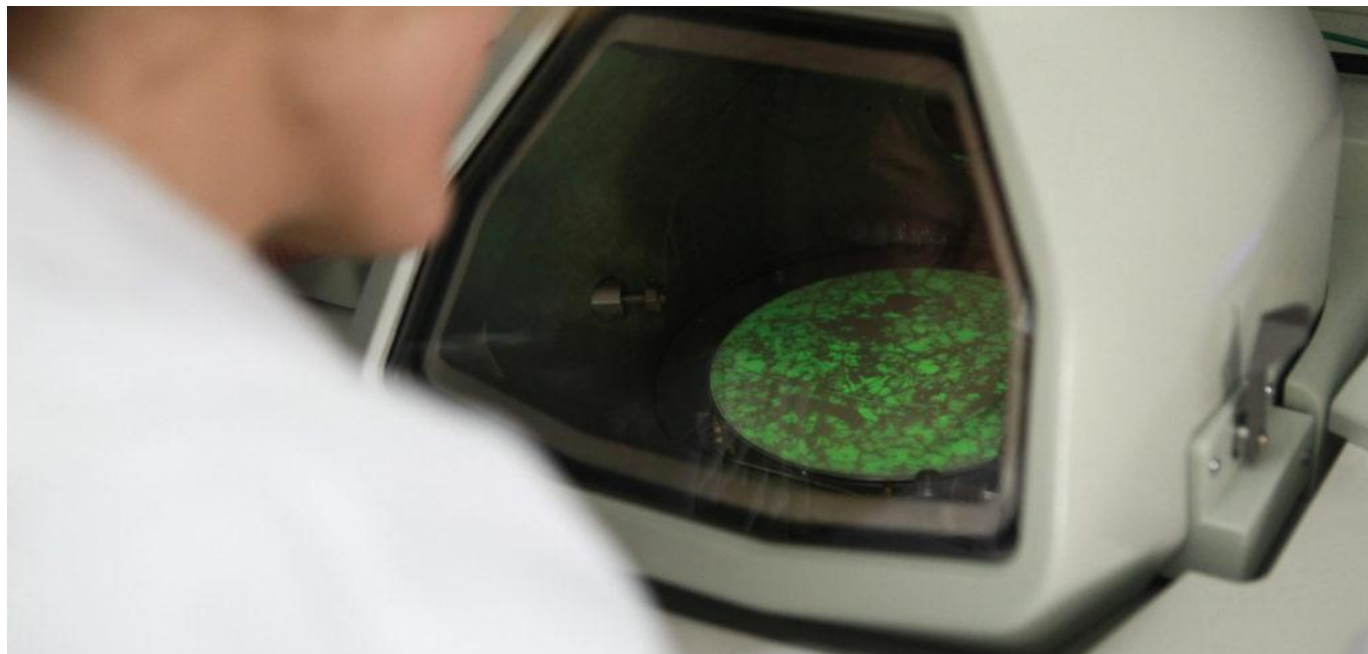
## Sketch of an objective lens



<https://www.jeol.com/words/emterms/20121023.060558.php#gsc.tab=0>

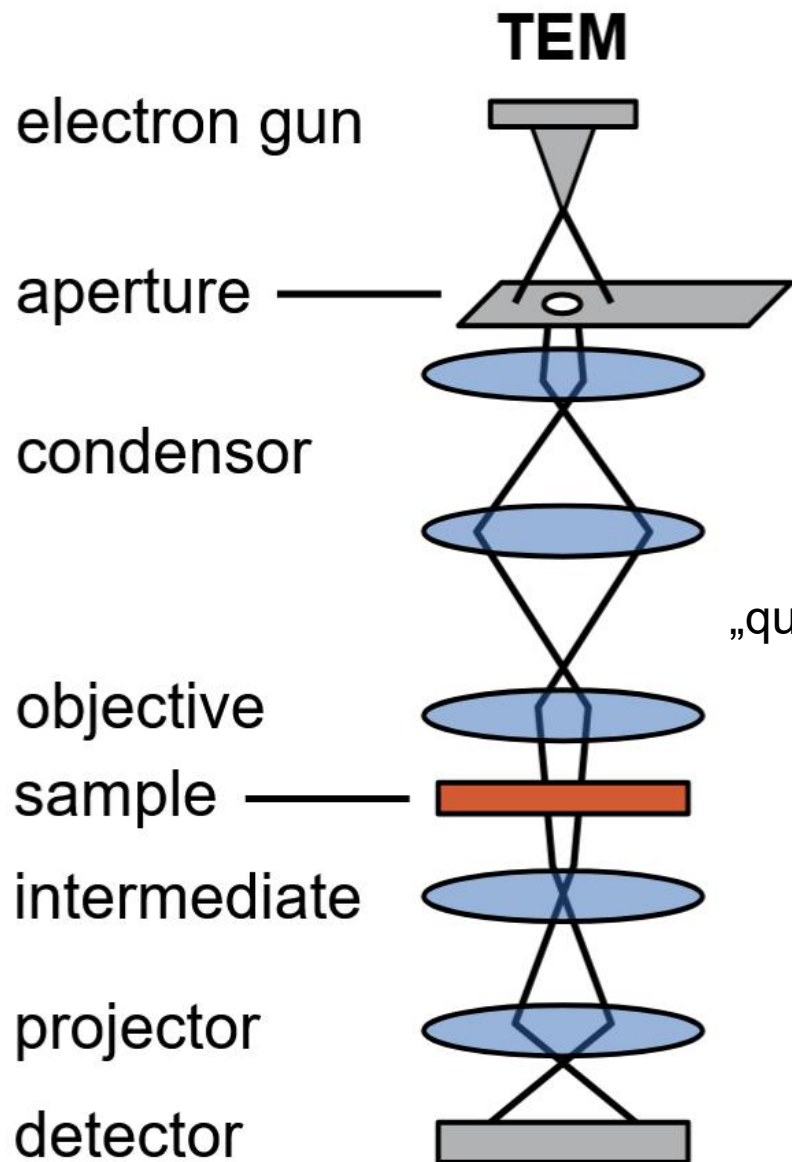


TEM image on a fluorescent screen while adjusting beam parameters

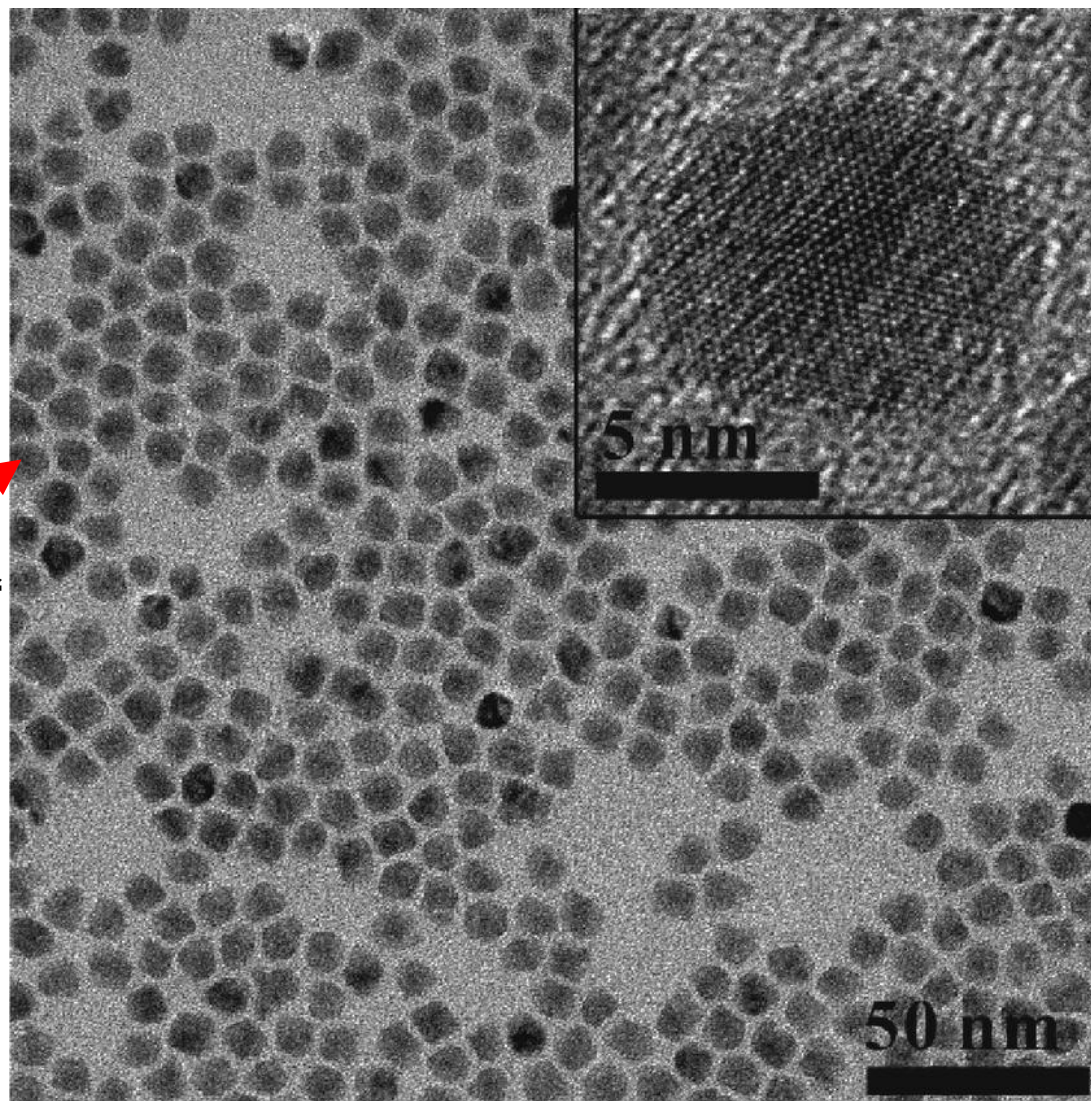


<https://www.itgagroup.com.au/asbestos-tem-analysis/>





„quantum dot“



DOI: 10.1109/TED.2017.2771785



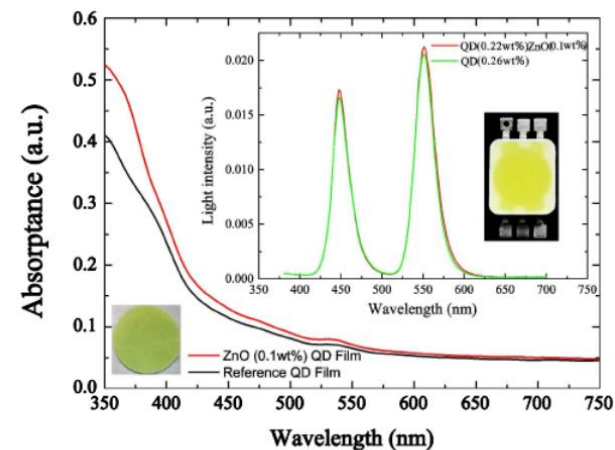
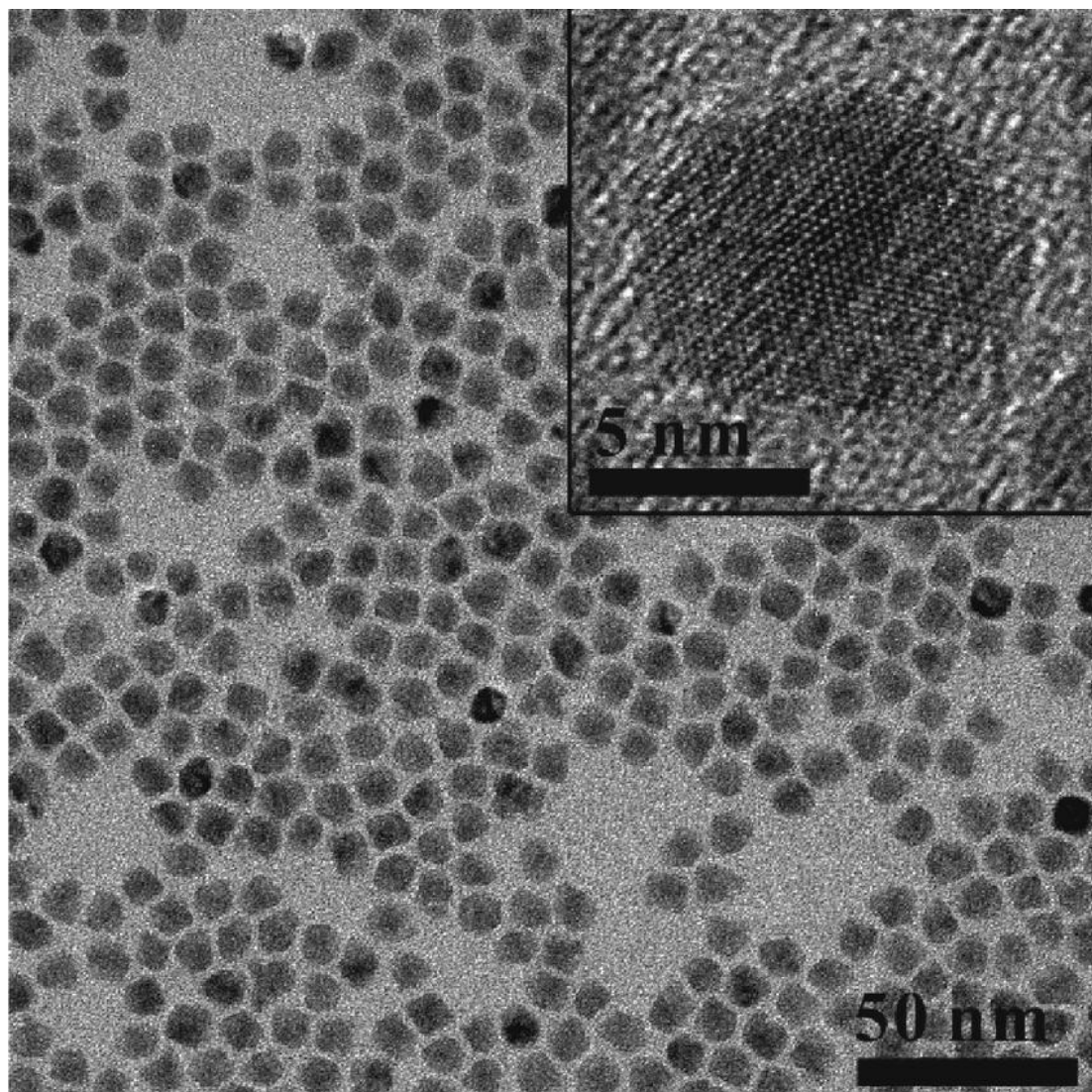


Fig. 8. Absorption spectra of the reference QD film (without ZnO) and the optimal QD film (0.1 wt% ZnO incorporated). The inset shows the emission spectra of 0.1 wt% ZnO-incorporated QD-LED (0.22 wt% QD), and none-ZnO QD-LED (0.26 wt% QD) under the similar color coordinate.

integrating sphere. In contrast to the radiation flux, the statistics show that the luminous flux initially increased with the

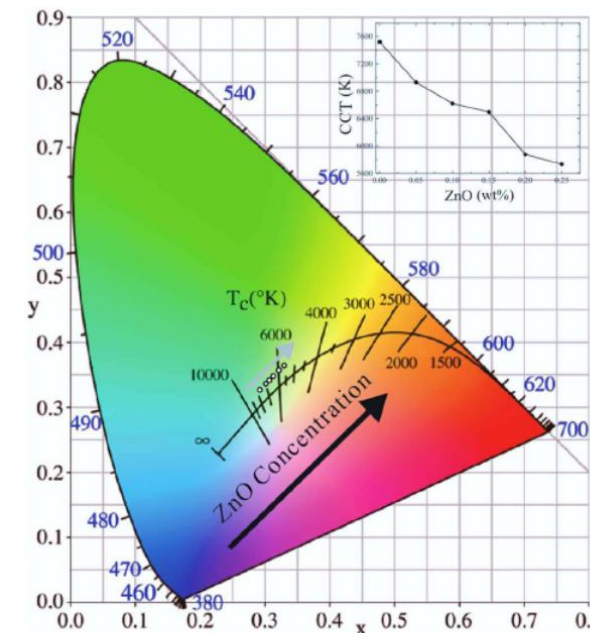
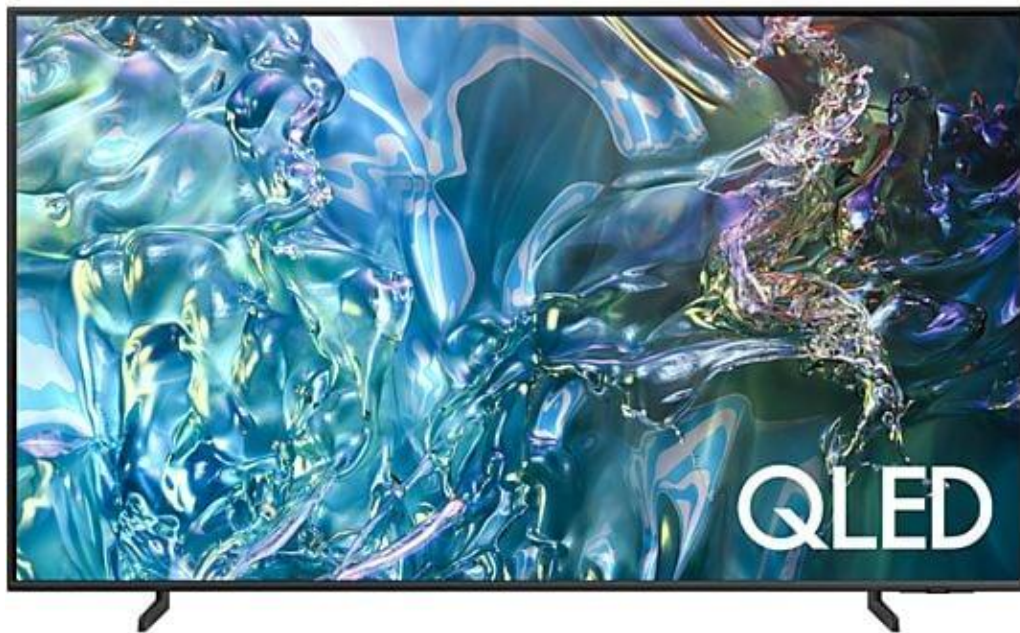


Fig. 9. Chromaticity coordinates and CCT of the QD-converted LEDs with different amounts of ZnO nanoparticles.

DOI: 10.1109/TED.2017.2771785



## Nobel Prize in Chemistry 2023



© Nobel Prize Outreach. Photo: Clément Morin

Mounji G. Bawendi

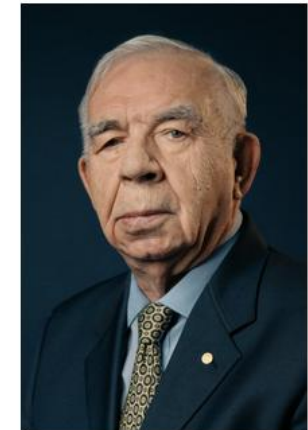
Prize share: 1/3



© Nobel Prize Outreach. Photo: Clément Morin

Louis E. Brus

Prize share: 1/3

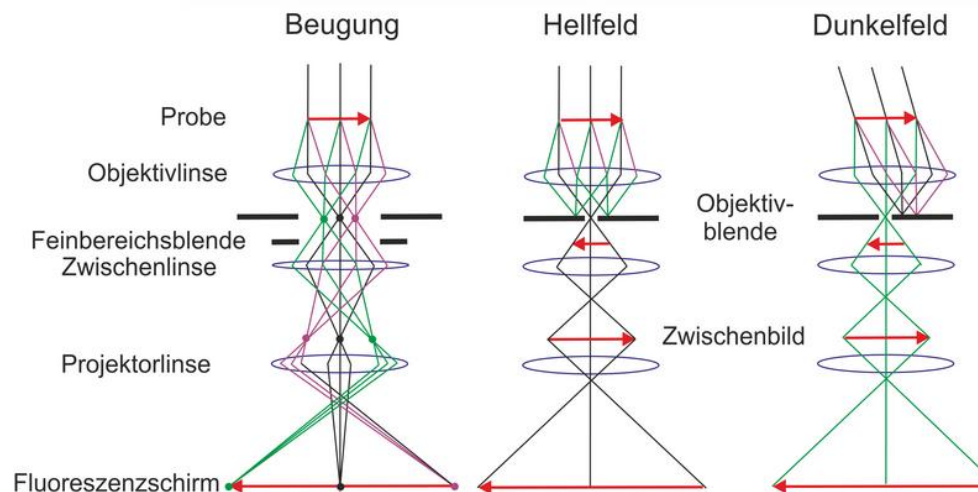
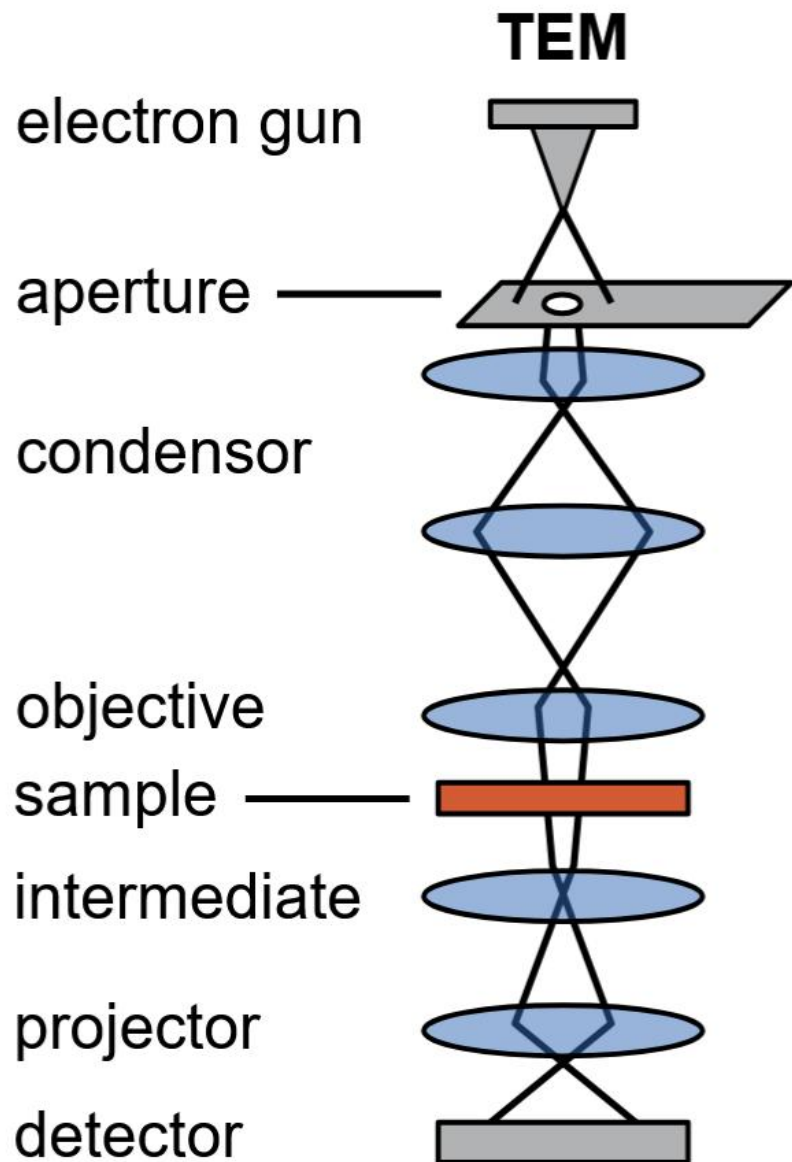


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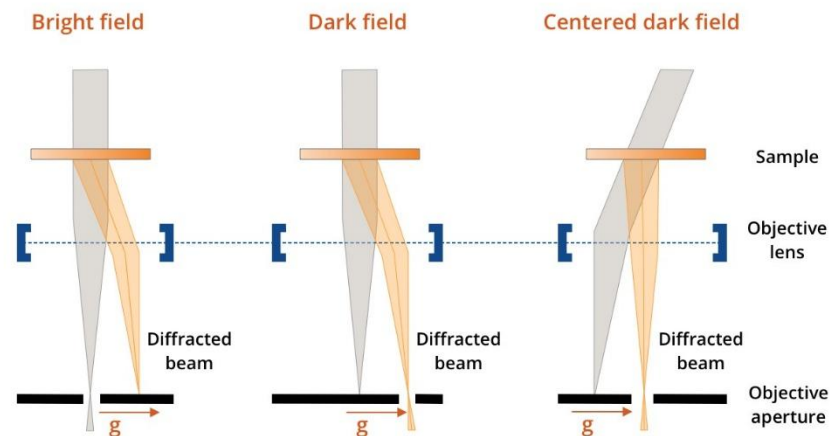
Aleksey Yekimov

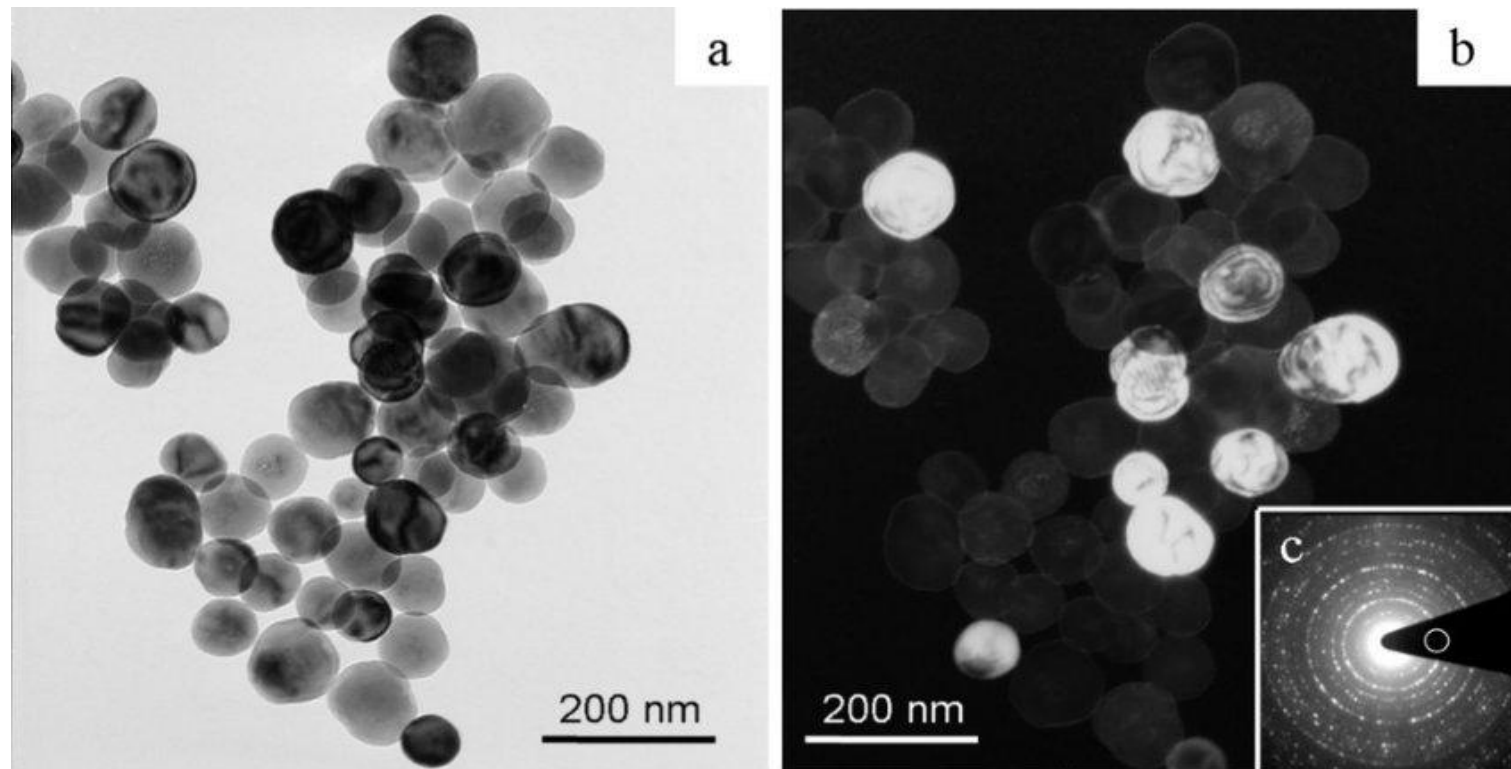
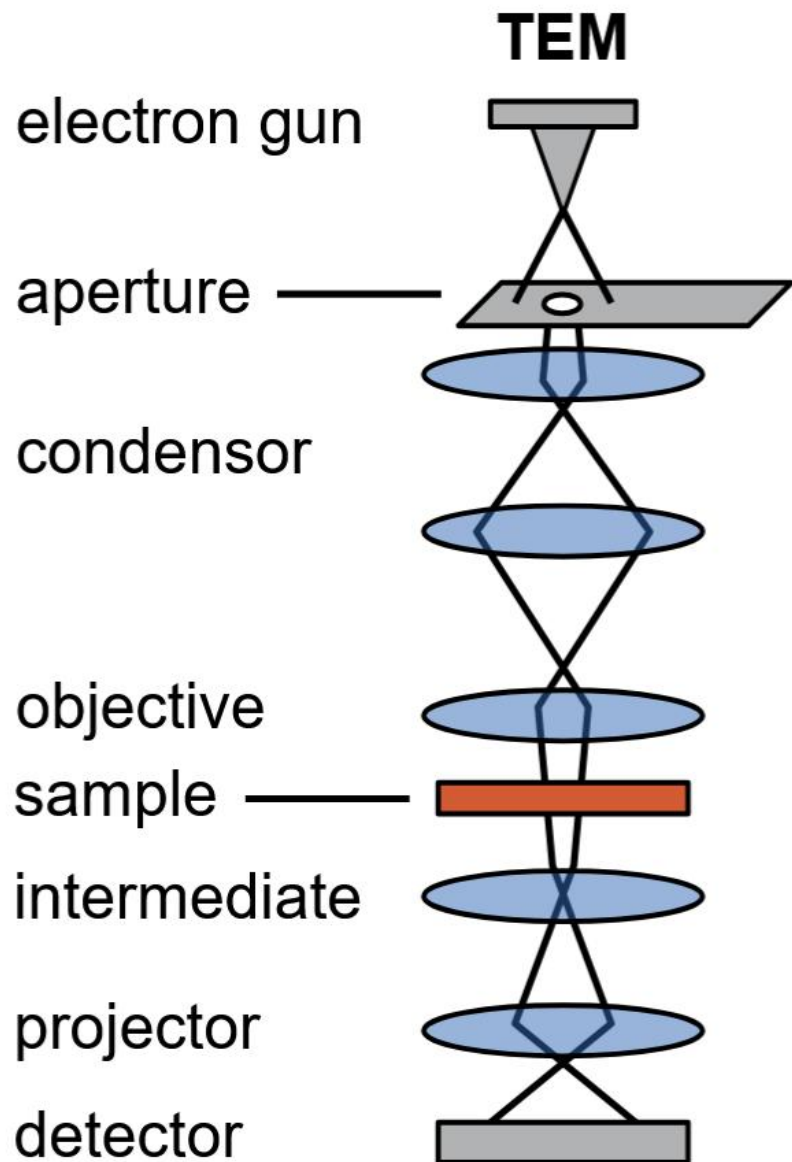
Prize share: 1/3

The Nobel Prize in Chemistry 2023 was awarded to Mounji G. Bawendi, Louis E. Brus and Aleksey Yekimov "for the discovery and synthesis of quantum dots"



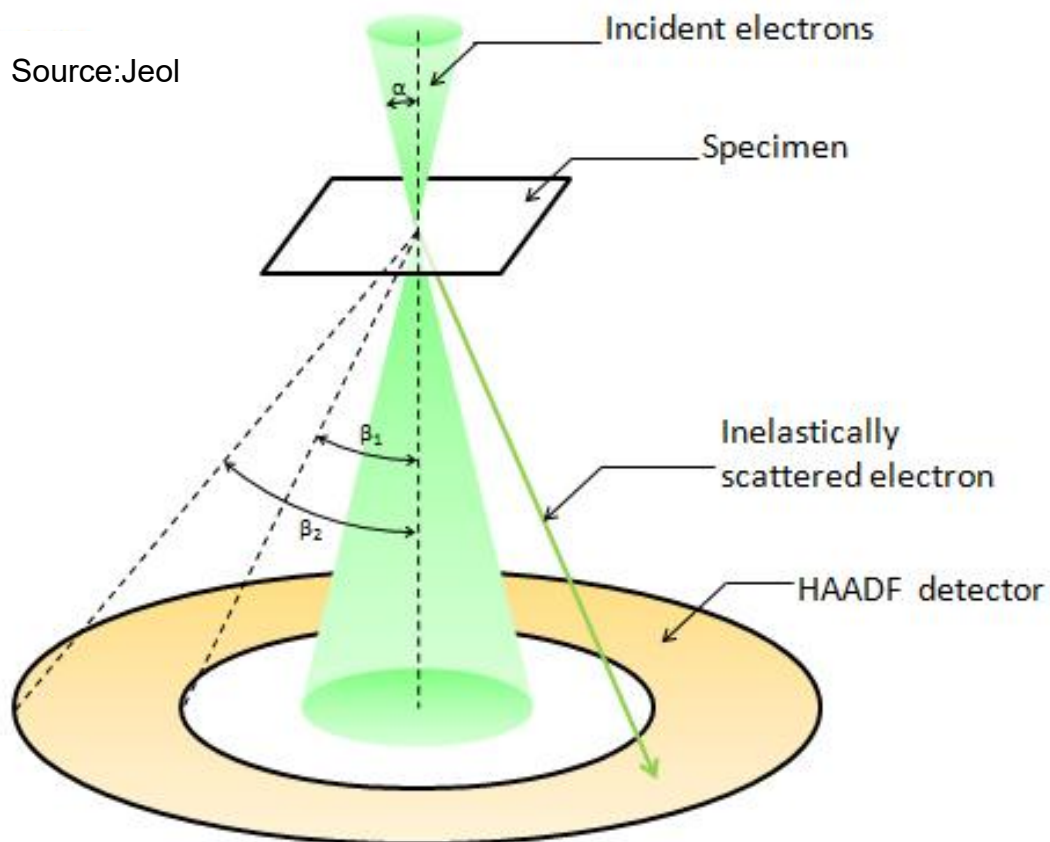
<https://physik.uni-paderborn.de/en/lindner/nanostrukturierung-nanoanalytik-photonische-materialien/research-1/temstem-2>



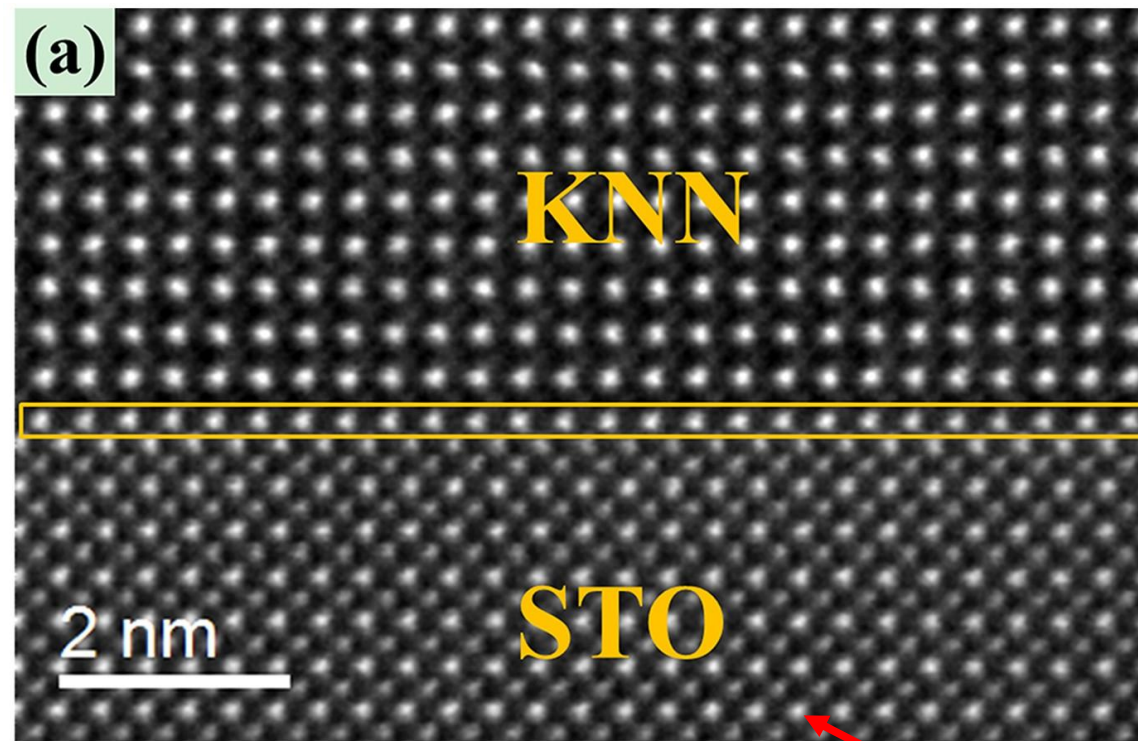


Source: Perovskite Oxide Nanocrystals — Synthesis, Characterization, Functionalization, and Novel Applications

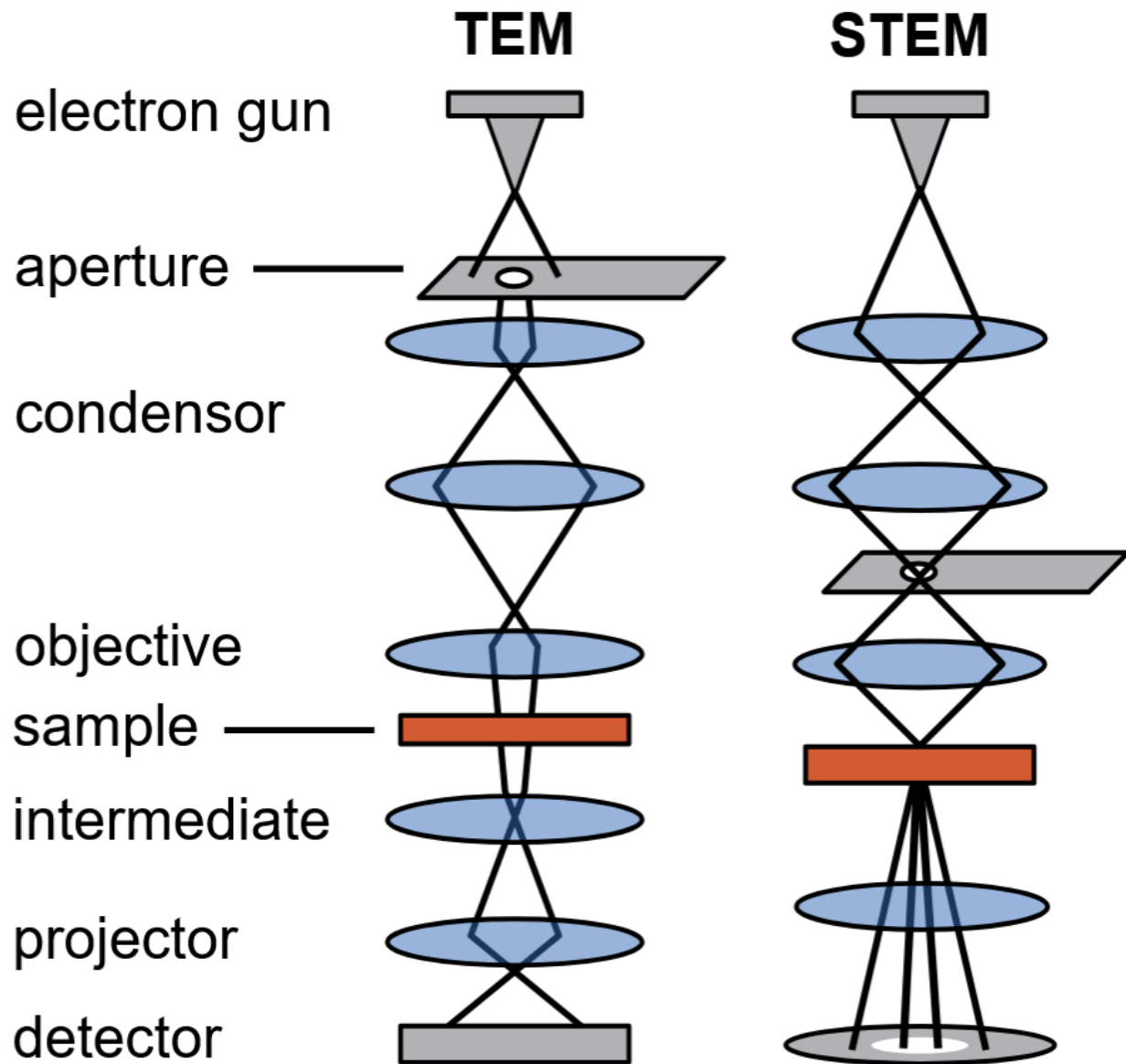




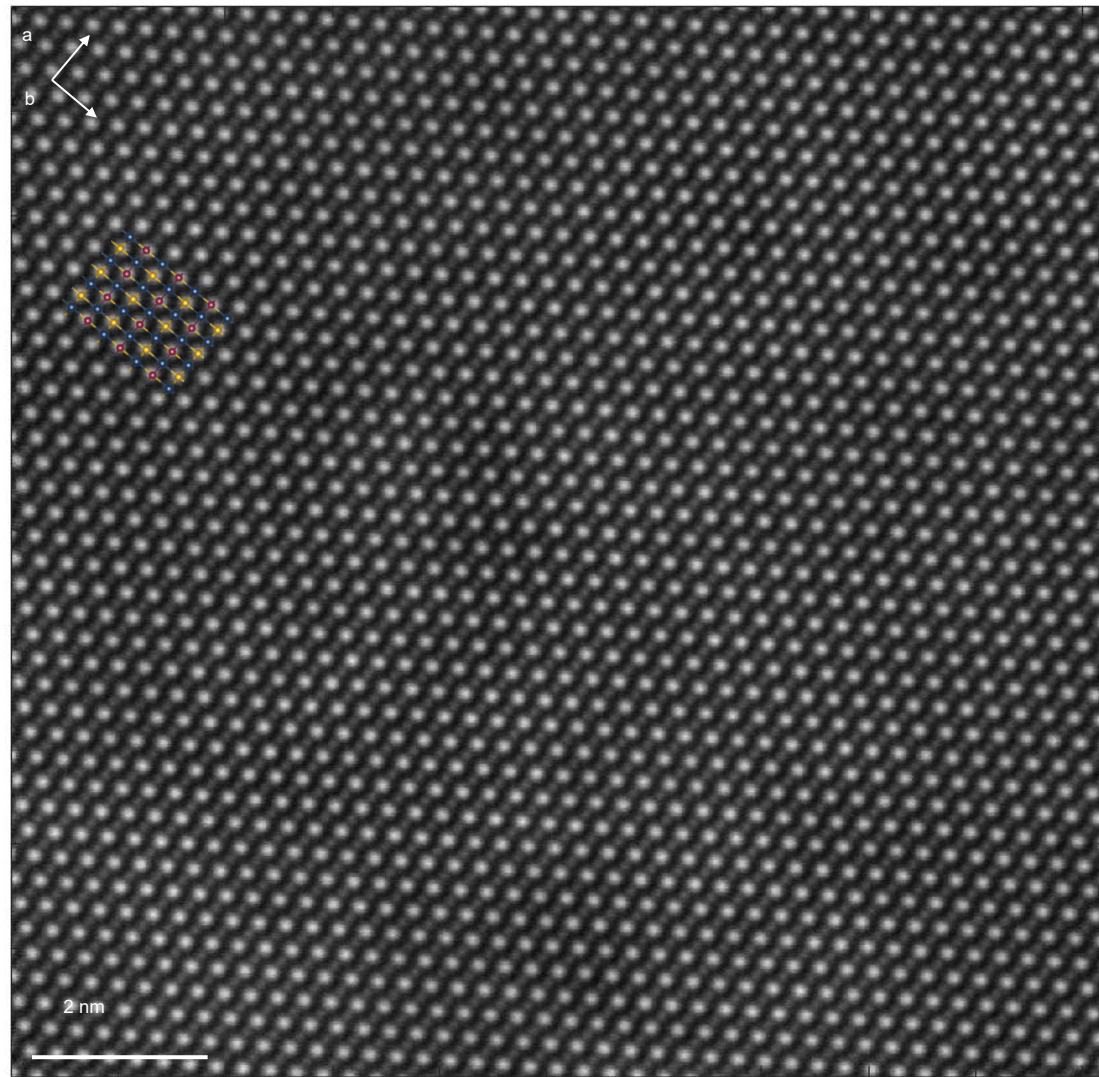
Bulk sample



<https://www.nature.com/articles/srep37788>

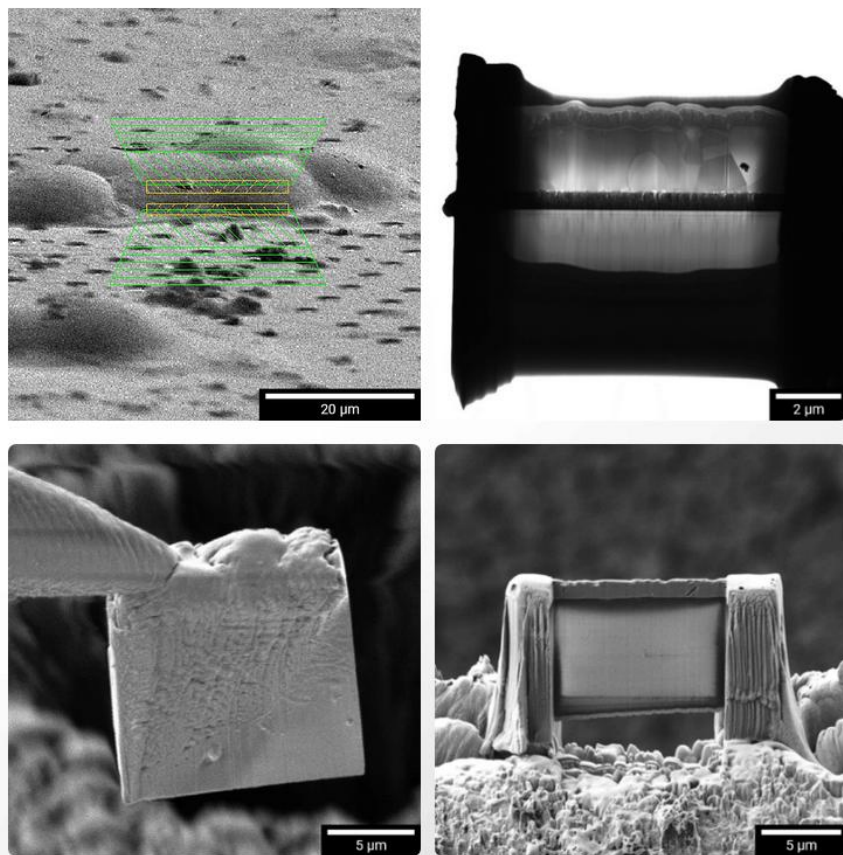


CrSBr – Layered antiferromagnet



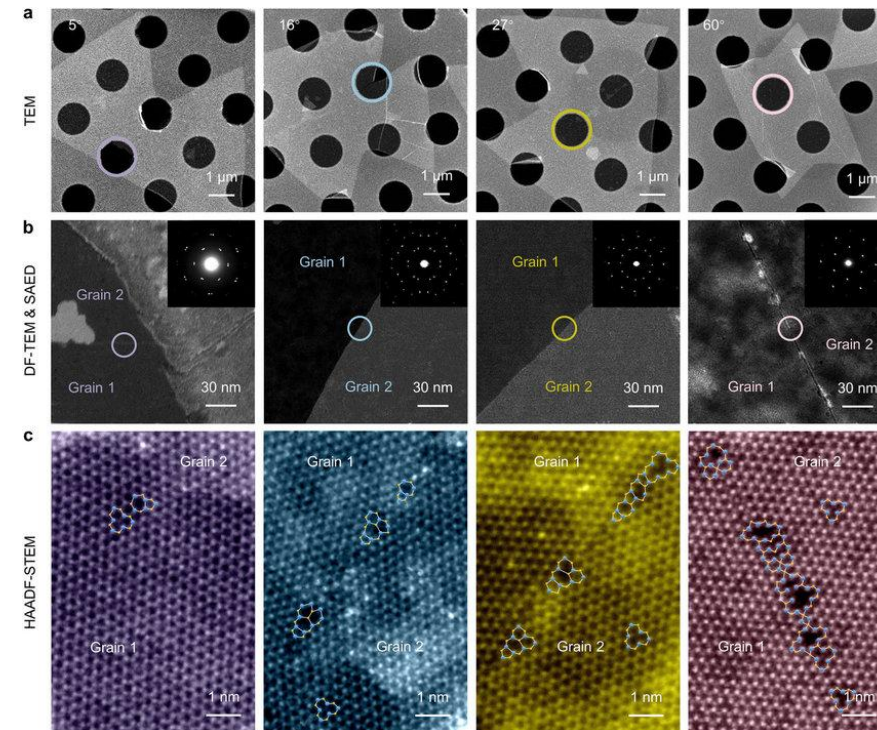


## Cut from bulk crystal

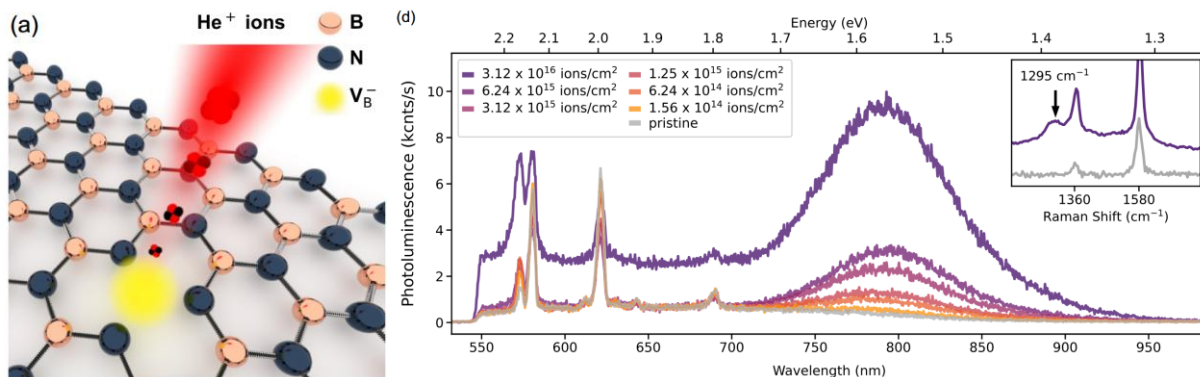


Source Tescan

## Make from layered crystal

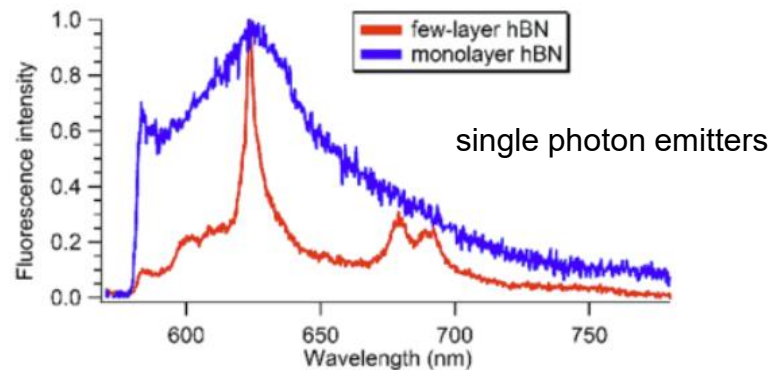


## Optically active defect centers for quantum sensing

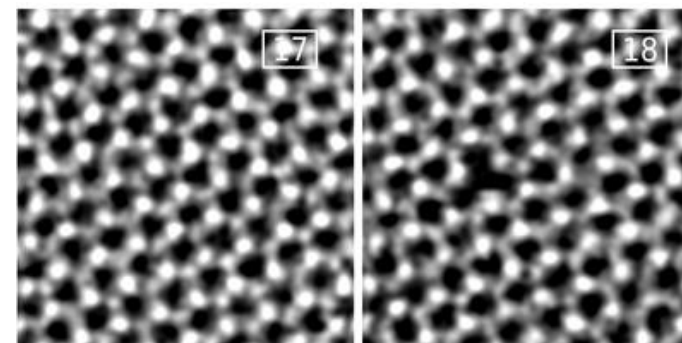


Phys. Rev. Materials 9, 056203 (2025)

## Optically active defect centers for quantum communication



## Knock-on damage



Small June 2023 19(39):e2301926 DOI:10.1002/sml.202301926

## Electron beam etching

